

From Police Workload to Auditable Performance Measures

Hurricane Ida, Public CAD, and the Measurement Frontier
for Reported DV-Related Calls

24 August 2026

Paper R12.2 Results R12

Abstract

Police workload models connect reported demand, effective capacity, prioritization, and service. Empirical use of those models also requires an observation layer that maps operational activity into auditable records. This paper develops that layer for public computer-aided dispatch (CAD) data and applies it to measurement of responsiveness, prioritization, and administrative follow-through for reported domestic-violence-related calls. Hurricane Ida supplies a system-level stress test of the shared public record. Its all-call window ranks first among 217 qualified reference windows and in two narrower designs; the dispatch–arrival field configuration also changes sharply over time and within recorded dispositions. These results establish that the released measurement product can become unstable under joint operational and record-production stress. A measure-specific identified set then maps each DV performance target to the evidence it requires. Public CAD directly supports declared-rule administrative counts and field completeness, while unresolved labels, missing endpoints, and broken lineage yield sharp bounds or selected summaries. Identified-set contraction and one-at-a-time necessity arguments characterize inclusion-minimal privacy-safe witness bundles: qualified labeling and retained aggregate lineage form the common core, followed by the clocks, priority histories, eligibility rules, and completion fields used by each measure. The framework turns workload concepts into a concrete data design for resource management and civilian oversight. The estimates remain system-level; the DV application defines an empirical agenda rather than a DV performance estimate.

Keywords: calls for service; police performance measurement; domestic violence; Hurricane Ida; measurement validity; partial identification

JEL codes: C18, C55, H12, K42

1 Introduction

Police departments allocate scarce capacity against changing reported demand. For domestic-violence-related calls, the practical questions are immediate: how long did calls wait, how were priorities revised, which operational endpoint was reached, and did the required downstream follow-up occur? The same quantities matter to civilian oversight because they locate delay and administrative attrition along the service chain.

Workload theory organizes the operational side of this problem. Let D_t^{CAD} denote reported demand entering CAD, C_t effective capacity, and P_t the priority process. The resulting service path is

$$S_t = g(D_t^{\text{CAD}}, C_t, P_t). \quad (1)$$

Neither true need nor attempted help-seeking is identical to this CAD denominator. The demand chain runs from latent need D_t^* through help-seeking attempts H_t and contacts received by the emergency system A_t to calls created in CAD, D_t^{CAD} . The paper studies performance conditional on that last population.

Public CAD is a tempting way to observe S_t : it is large, timely, and granular. Yet every released timestamp, priority, and disposition also passes through a recording, retention, and export process R_t . Researchers observe

$$Y_t = h(S_t, R_t), \quad (2)$$

and a candidate performance measure is $\theta_k = m_k(S)$. The relevant identified set contains the values of θ_k generated by operational and record-production paths compatible with the released data and admitted witnesses. Its width determines whether a measure is directly supportable, bounded, selected, or unavailable.

Hurricane Ida establishes why this observation layer matters empirically. Access to 911, police operations, and record production were strained together. The analysis asks whether relationships among released fields remained within a frozen reference class. It finds a reference-extreme, time-structured, within-disposition change in the public trace. The stress test cannot assign that change to service or recording, which is precisely the ambiguity that a performance measure must resolve.

The paper’s central argument is:

The paper develops a measurement-admissibility framework using Hurricane Ida as a system-level stress test, then applies it to the workload and follow-through measures needed for reported DV-related calls.

Three connected contributions implement this argument. First, the empirical module constructs four dispatch–arrival field-presence states, standardizes them across ten frozen twelve-hour bins, and compares Ida with 217 qualified reference windows. Second, the identification module maps each performance target into the set of service paths consistent with the public record. Third, the DV application follows classification, response endpoints, priority histories, EPR production, and follow-up to characterize inclusion-minimal aggregate witness bundles for each target.

Each element has a distinct role. Existing research develops police-performance targets, workload relationships, administrative-data error, and partial-identification methods. The integrated contribution is an auditable bridge among them: an empirical diagnostic for the shared CAD

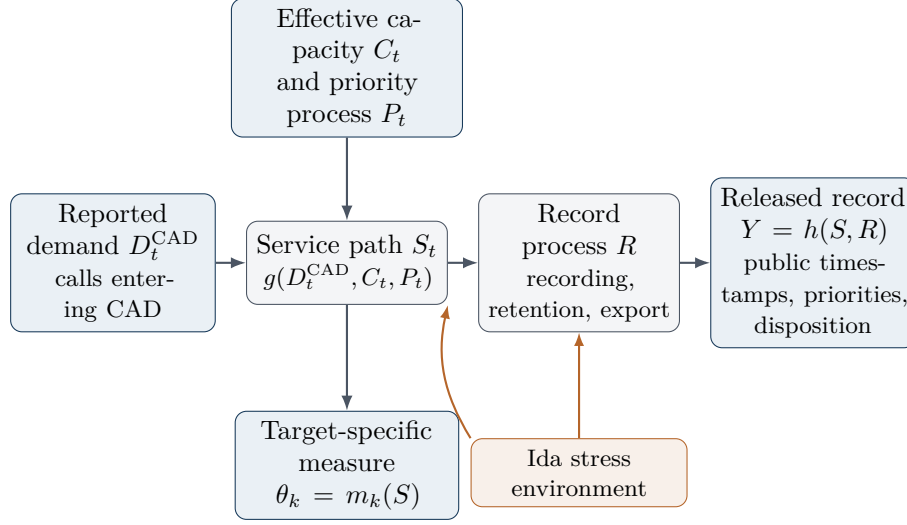


Figure 1. Workload produces the service path; the record process produces the public observation. Target-specific witnesses establish which service measures the observation supports.

observation layer, target-specific identified sets for a downstream crime category, and privacy-safe release specifications whose components are justified by necessity arguments. This extends the police-workload perspective associated with Beland (2026). Reported demand and effective supply shape service, while the administrative observation layer determines which workload implications can be measured.

The result is constructive for both management and oversight. Better lineage, clocks, and histories contract identified sets directly. They can support future allocation models and sharper accountability measures on a common reported-call denominator. Estimating a DV performance level, the effect of a reform, or a behavioral mechanism remains a subsequent empirical task with its own classification authority and research design.

2 Economic and empirical background

Police performance is multidimensional, and administrative convenience is a weak basis for choosing a metric (Moore and Braga, 2004; Tiwana et al., 2015). More generally, public-sector metrics can redirect effort toward recorded targets and alter the records used for accountability (Courty and Marschke, 1997; Eckhouse, 2022). For domestic and family violence, Rollings and Taylor (2008) propose measures of recording accuracy, repeat attendance, repeat offending, charging, information supplied to responders, and victim satisfaction. EIGE likewise organizes police and justice data around thirteen IPV indicators and comparable administrative definitions (European Institute for Gender Equality, 2023). These studies define substantive targets. The present paper asks which targets can be recovered from a public CAD release.

Responsiveness illustrates the issue. Faster police arrival can affect clearance and injury outcomes (Blanes i Vidal and Kirchmaier, 2018; DeAngelo et al., 2023). Yet response time varies across reporting, dispatch, travel, and combined clocks, and the literature uses heterogeneous priority definitions and call populations (Verlaan and Ruiter, 2023). Rich internal data can link staffing, call volume, proactive work, priority, and response times (Mourtgos et al., 2024). A public release that omits the queue, availability, or an endpoint supports a narrower statistic.

The DV literature further separates reported demand from the underlying phenomenon. Calls, incidents, arrests, hotline contacts, and hospital records can move differently during the same disruption (Leslie and Wilson, 2020; Bullinger et al., 2021; Miller et al., 2022, 2024). Police composition can affect reporting (Miller and Segal, 2019), while stable identifiers permit linkage from police reports to later economic outcomes (Adams et al., 2024). This paper therefore targets performance conditional on a reported DV-related call entering CAD.

Administrative records reflect selection, classification, and linkage. Reporting, coding, and processing create successive gates (Klinger and Bridges, 1997; Boivin and Cordeau, 2011; Boivin and Ouellet, 2014; Simpson and Orosco, 2021; Knox et al., 2020). Call-takers assign categories and priorities before resources are deployed (Pandey, 2025). Fragmented DV systems and inconsistent identifiers complicate links among incidents, reports, referrals, and follow-up (Phoenix, 2023; Vomfell and Povala, 2023). Changes in incentives can alter registered crime (Ahmed et al., 2026). The measurement design must therefore describe how an operational stage becomes an observed record.

Econometric work provides the corresponding identification logic. Classical identification asks whether the observable distribution distinguishes the underlying structure (Koopmans and Reiersøl, 1950; Rothenberg, 1971; Lewbel, 2019). Partial identification retains the complete set of observationally equivalent target values when it does not (Manski, 1989, 1990, 2003; Molinari, 2020). Measurement error in nonlinear settings generally requires auxiliary measurements, instruments, validation data, or explicit restrictions (Chen et al., 2011). Administrative sources can measure a different concept or contain their own linkage and classification errors (Kapteyn and Ypma, 2007). Dominitz and Sherman (2006) derive sharp bounds on school-performance measures when an administrative verification indicator is imperfect; their measure-specific treatment of validity is especially close to the present application. Communicating the resulting nonsampling and conceptual uncertainty is part of valid official measurement (Manski, 2015).

Two further connections organize the witness analysis. Blackwell (1953) compares information structures by the decisions they support, while Masten and Poirier (2020) locates boundaries at which conclusions cease to survive relaxed assumptions. Here, each administrative witness refines the feasible set of operational paths for a declared performance target. Sufficiency is reached when the target’s identified set meets a prespecified precision; one-at-a-time removal establishes whether every retained witness is necessary. The resulting inclusion-minimal bundle is a target-specific release design rather than a universal ranking of data fields.

Economists also use disasters to reveal system dependencies that routine variation leaves hidden. The Great East Japan Earthquake exposed propagation through directly observed firm networks (Carvalho et al., 2021); hurricanes and outages reveal payment resilience by combining transactions, outages, surveys, and experimental evidence (Alvarez et al., 2026). Environment comparisons can also test whether a statistical relationship remains invariant, although causal interpretation requires additional structural assumptions (Peters et al., 2016). BELAND-PLUS uses Ida at the diagnostic stage: the event tests stability of the released record, while operational witnesses determine the performance measure.

Partial identification supplies the formal response. Missing outcomes yield sharp regions, restrictions on misclassification can tighten them, and additional evidence or restrictions contract the set of observationally equivalent values (Manski, 1990, 2003; Horowitz and Manski, 1995; Molinari, 2008, 2020; Tamer, 2010). BELAND-PLUS applies that logic to the measurement stage itself. Ida

establishes the empirical relevance of the problem; the DV chain defines the candidate functionals and the witnesses that contract their identified sets.

3 Institutional setting and research design

3.1 Hurricane Ida as a system-level stress test

Hurricane Ida made landfall in Louisiana on 29 August 2021. The Office of the Independent Police Monitor documented twelve-hour police shifts, a period when 911 dispatch was down, anti-looting deployments, and a citywide curfew during the event (Office of the Independent Police Monitor, 2021). Reporting, police operations, and record production were therefore exposed to simultaneous stress. This is precisely what makes Ida informative as a measurement test: it can reveal whether relationships among public fields remain stable when the system producing them is under unusual strain.

Ida is the test bed. The design uses its documented operational conditions to probe the public measurement layer; the observed topology remains compatible with several operational and record-production pathways.

3.2 Why an all-call stress test informs DV measurement

The identified-set logic can be written from the schema alone. Ida adds the empirical premise that makes the exercise consequential: the shared public observation layer actually moved outside its frozen reference class under system stress. The result replaces a generic warning about administrative data with a measured failure of stability in the fields later used for crime-specific performance analysis.

The empirical analysis uses all reported calls because every downstream call-type measure inherits that CAD layer. A DV labeling rule changes the analytic population, while its priorities, timestamps, dispositions, and links still pass through the tested record-production system. The transfer is therefore a measurement requirement. Each DV measure must establish its denominator, labeling rule, clock, endpoint selection, and lineage. The system-level estimates supply no DV-specific magnitude.

3.3 Making the public record’s shape measurable

The analysis uses the official 2021 New Orleans Calls for Service file (City of New Orleans, 2021). The denominator is the frozen public-created-call universe used throughout the project. For record i , define

$$D_i = \mathbf{1}\{\text{a valid public dispatch field is present}\}, \quad (3)$$

$$A_i = \mathbf{1}\{\text{a valid public arrival field is present}\}. \quad (4)$$

The two indicators generate four mutually exclusive public record states:

Exactly one state is realized for every record, so the state shares sum to one. This simplex identity is central: when the share of J_{11} falls, the mass must move to one or more other public states. The analysis is therefore about redistribution within the public record, not simply about a count declining.

Table 1. Public field-presence states

State	Dispatch field	Arrival field	Safe interpretation
J_{00}	absent	absent	neither public field is present
J_{01}	absent	present	arrival field is present; dispatch field is absent
J_{10}	present	absent	dispatch field is present; arrival field is absent
J_{11}	present	present	both public fields are present

These states record public field presence. Physical-stage status requires a separate operational bridge.

3.4 Event window and time bins

The Ida window runs from 29 August 2021 at 00:00 through 3 September at 00:00 local time. It is divided prospectively into ten twelve-hour bins. To avoid unexplained internal labels, the paper reports calendar time directly; the technical labels are retained only for reproducibility.

Table 2. Calendar interpretation of the focal bins

Technical bins	Calendar interval	Role in the descriptive result
B1	29 Aug., 00:00–12:00	opening interval; not among the nine threshold-exceeding cells
B2–B5	29 Aug., 12:00–31 Aug., 12:00	earlier pattern: lower J_{11} share
B6–B7	31 Aug., 12:00–1 Sep., 12:00	later pattern: lower J_{11} and higher J_{01}
B8	1 Sep., 12:00–2 Sep., 00:00	negative J_{11} tail
B9–B10	2 Sep., 00:00–3 Sep., 00:00	no cell exceeds the common threshold

3.5 Standardized event–reference contrasts

Within stratum s , time bin t , and state j , let p_{jst}^E and p_{jst}^R denote the event and reference shares. Using the frozen reference-derived common-support weights ω_{st}^R , the standardized event–reference contrast is

$$\Delta_{j,t} = \sum_{s \in \mathcal{S}_t} \omega_{st}^R (p_{jst}^E - p_{jst}^R), \quad \sum_{s \in \mathcal{S}_t} \omega_{st}^R = 1. \quad (5)$$

Because the four state shares sum to one,

$$\Delta_{00,t} + \Delta_{01,t} + \Delta_{10,t} + \Delta_{11,t} = 0. \quad (6)$$

The primary topology matrix keeps the three independent coordinates $(\Delta_{01,t}, \Delta_{10,t}, \Delta_{11,t})$ for the ten bins.

3.6 Benchmarking rarity against frozen reference windows

The topology score measures how closely any authorized restricted-support representation can approximate the full three-coordinate path while respecting its support envelope. Let G be the 10×3 contrast matrix and $\mathcal{X}$ the frozen finite set of authorized support triples. The full score can be written compactly as

$$U_{\text{full}}(G) = \min_{X \in \mathcal{X}} \max_{j \in \{01,10,11\}} \min_{\beta_j} \|g_j - X\beta_j\|_{\infty}, \quad (7)$$

subject to the rowwise capacity-envelope restrictions defined in the mathematical appendix. A larger score means that the best authorized representation leaves a larger maximum cell discrepancy. The score is a geometric measure of public-record fit.

The same estimator is applied to every qualified reference window. We report empirical rank and the add-one reference fraction $r/(R+1)$ as descriptive measures of reference rarity.

4 Empirical results

The stress test must establish more than an unusual average. It asks whether Ida lies outside the range of the same measurement applied to qualified reference windows, whether the public-record shape changes during the event, and whether the later movement is merely a change in recorded-disposition mix. The three results below answer those questions in sequence.

4.1 Ida is reference-extreme

Ida’s locked score interval is

$$U_{\text{full}}(G_{\text{Ida}}) \in [0.250734484671, 0.250734494671]. \quad (8)$$

No reference interval overlaps it. Ida ranks first among 217 full qualified windows, first among 153 stage-era-matched windows, and first among 86 same-season-stage windows. The corresponding add-one reference fractions are $1/218$, $1/154$, and $1/87$.

The largest reference upper endpoint is approximately 0.2125, leaving a minimum interval separation of about 0.0382 from Ida. This separation is more informative than the narrower fact that Ida lies just above the frozen M7A restricted-support exclusion threshold of 0.25. The threshold is a prespecified model-fit boundary; the empirical rank comparison is invariant to it.

4.2 The public-state configuration changes after midday on 31 August

A cell is classified as unusually large when its absolute contrast exceeds the 95th percentile of the reference-window maximum absolute cell contrast,

$$c_{0.95} = 0.209397476359. \quad (9)$$

Nine bin-state cells exceed this common threshold. The pattern is easier to understand in calendar time than through technical bin labels:

- From the afternoon of 29 August through the morning of 31 August, the abnormal cells are reductions in J_{11} , the state in which both public fields are present.
- From the afternoon of 31 August through the morning of 1 September, J_{11} remains sharply lower, while J_{01} —arrival present but dispatch absent—becomes sharply higher.
- During the afternoon and evening of 1 September, the negative J_{11} contrast remains above the threshold, but J_{01} no longer does.

The result is a redistribution statement: one public state falls while another rises. Its generating pathway remains set-valued.

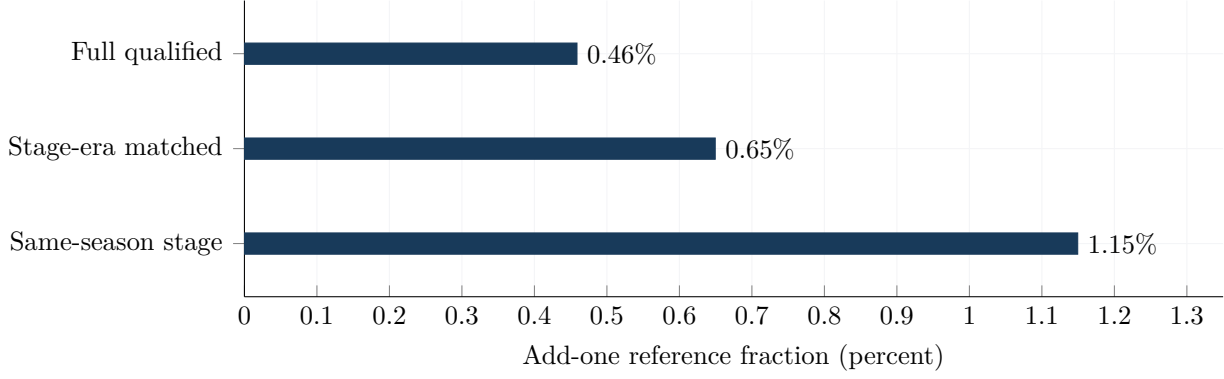


Figure 2. Ida’s position in the three qualified same-estimator reference designs. The plotted fractions describe empirical reference rarity.

J_{01} : arrival present, dispatch absent	–	–	–	–	+0.307	+0.323	–
J_{11} : both fields present	-0.422	-0.354	-0.216	-0.266	-0.507	-0.455	-0.228
	Aug 29 PM	Aug 30 AM	Aug 30 PM	Aug 31 AM	Aug 31 PM	Sep 1 AM	Sep 1 PM

Figure 3. The nine threshold-exceeding public-record cells. Values are event-minus-reference probability-point contrasts; a dash marks a cell below the common threshold. The figure describes public field configurations.

4.3 The later shift occurs within recorded dispositions

The next question is whether the late pattern could arise solely because the mix of dispositions changed. Restrict attention to arrival-observed records, $J_{01} + J_{11}$. For disposition r and population $z \in \{E, R\}$, let

$$q_r^z = \Pr(D = 1 \mid A = 1, R = r, z), \quad \pi_r^z = \Pr(R = r \mid A = 1, z). \quad (10)$$

The aggregate dispatch-field share among arrival-observed records is $q_{\text{all}}^z = \sum_r \pi_r^z q_r^z$. The exact symmetric Kitagawa identity is

$$\begin{aligned} q_{\text{all}}^E - q_{\text{all}}^R &= \sum_r \frac{q_r^E + q_r^R}{2} (\pi_r^E - \pi_r^R) \\ &\quad + \sum_r \frac{\pi_r^E + \pi_r^R}{2} (q_r^E - q_r^R). \end{aligned} \quad (11)$$

The first term is the recorded-disposition composition component. The second is the within-disposition component.

For 31 August afternoon/evening, the total contrast is -0.4950 , the composition component is $+0.0119$, and the within-disposition component is -0.5069 . For 1 September morning, the corresponding values are -0.4824 , $+0.0123$, and -0.4946 . The within-disposition component is therefore responsible for the direction and almost all of the magnitude of the aggregate shift.

The sign pattern strengthens the accounting result. For Gone on Arrival (GOA), Report to

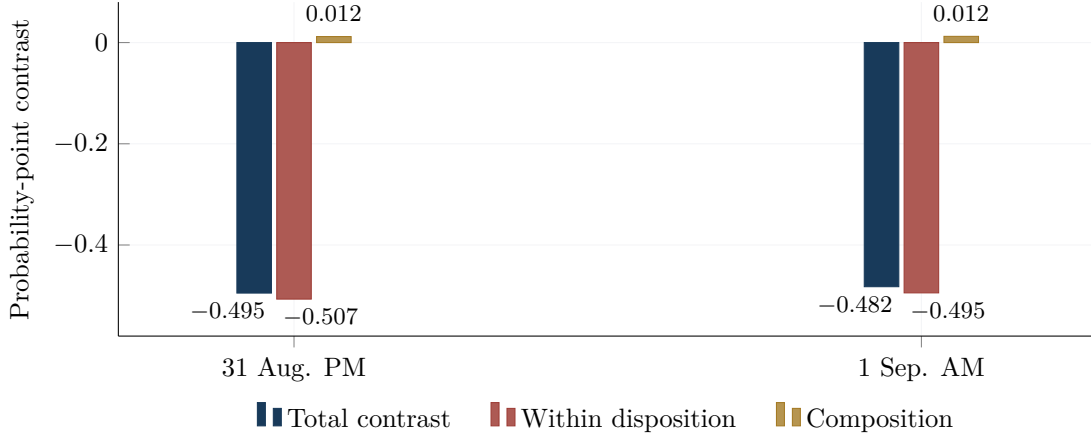


Figure 4. Exact accounting decomposition of the late aggregate shift. The small positive composition term has the opposite sign from the negative total contrast.

Follow (RTF), and Necessary Action Taken (NAT), $\Delta H_{01,r} > 0$ while $\Delta H_{11,r} < 0$ in both late bins. A composition-only model requires matching signs, so the six prespecified cells rule out that explanation alone. The remaining pathways form an identified set.

Table 3. Within-disposition change in the public dispatch-field share among arrival-observed records

Calendar bin	GOA	RTF	NAT
31 Aug., 12:00–24:00	−0.440	−0.567	−0.532
1 Sep., 00:00–12:00	−0.265	−0.459	−0.649

Each value is an event-minus-reference contrast in the public dispatch-field share among arrival-observed records within the named recorded disposition. GOA is Gone on Arrival; RTF is Report to Follow; NAT is Necessary Action Taken. All three are administrative categories.

Table 3 reports conditional-share contrasts, whereas Appendix Section 5.1 factors the corresponding joint masses as $H_{11} = mq$ and $H_{01} = m(1 - q)$; Appendix Table 4 reports their contrasts. The tables therefore describe linked but distinct quantities: the first localizes conditional public-field movement, while the second supplies the sign witnesses used to exclude composition-only.

Together, the results establish an extreme, time-structured change in the public administrative trace, including movement within recorded dispositions. The stress test therefore rejects routine interpretation of the released fields as a mechanically stable window onto operations during Ida.

This is the empirical bridge to DV measurement. A DV label selects records after they have passed through the same field-production system. The performance question then becomes measure-specific: which operational functionals retain a unique interpretation after that selection?

The all-call result neither estimates DV demand nor implies that every DV measure fails. It establishes a common observation-layer problem. The DV application determines which targets survive that problem and which require additional evidence.

Stress-test implication. Ida establishes instability in the released measurement product. Operational interpretation requires a separate identified-set analysis for each downstream

measure.

5 From the Ida diagnostic to measure-specific identification

The empirical statistic and the performance target are different mathematical objects. The stress-test statistic is a function of released records,

$$T(Y) := U_{\text{full}}(G(Y)). \quad (12)$$

Its reference rank establishes that Ida’s public-field path is unusual within the frozen comparison design. A candidate performance target instead depends on the latent operational path S .

Let W denote the available semantic and operational witnesses, and let $\mathcal{C}(W)$ collect the paths and recording processes consistent with those witnesses. For candidate measure $\theta_k = m_k(S)$, define

$$\mathcal{I}_k(Y, W) = \{m_k(S) : Y = h(S, R), (S, R) \in \mathcal{C}(W)\}. \quad (13)$$

The measure is directly supportable when this set is a singleton, bounded when it is informative but non-singleton, selected when its observed version conditions on a narrower risk set, and unavailable when current evidence leaves the target unrestricted or undefined. New witnesses contract $\mathcal{C}(W)$ and therefore weakly contract $\mathcal{I}_k(Y, W)$.

5.1 Internal stage occurrence versus public capture

Suppose future semantic evidence established a valid direct bridge between a qualified internal administrative dispatch event and the public dispatch field. Let

$$q^{\text{pub}} = \Pr(D_{\text{pub}} = 1 \mid A_{\text{pub}} = 1, R = r, t), \quad (14)$$

$$u = \Pr(U_{\text{dispatch}} = 1 \mid A_{\text{pub}} = 1, R = r, t), \quad (15)$$

$$\chi = \Pr(D_{\text{pub}} = 1 \mid U_{\text{dispatch}} = 1, A_{\text{pub}} = 1, R = r, t). \quad (16)$$

Under exact nesting, the observed public share factorizes as

$$q^{\text{pub}} = u\chi. \quad (17)$$

For any $0 < q^{\text{pub}} < 1$, the identified set is

$$\mathcal{I}(u, \chi \mid q^{\text{pub}}) = \{(u, q^{\text{pub}}/u) : u \in [q^{\text{pub}}, 1]\}. \quad (18)$$

The same public value is compatible with perfect public capture and a lower internal-stage prevalence, complete internal-stage prevalence and imperfect public capture, and every intermediate combination.

The product identifies a curve rather than a unique pair. Joint public–internal counts would separate internal-stage prevalence from public capture. This example is a special case of [equation \(13\)](#): timestamp precision does not establish the mapping from a public field to the operational stage it is intended to measure.

A DV labeling rule restricts the analytic population but leaves the observation map unchanged. The DV results below are derived by applying [equation \(13\)](#) to that selected chain. They are

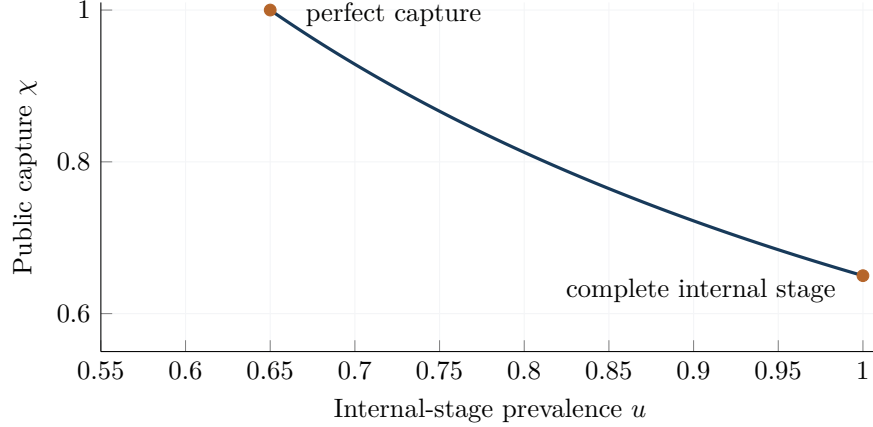


Figure 5. Illustrative product identified set for $q^{\text{pub}} = 0.65$. Observing the product selects a curve of internal-stage and capture values. The value 0.65 is illustrative.

measurement results, not an empirical DV subset analysis.

5.2 Queue, availability, priority history, and continuity

Queue dynamics live on a different denominator. A qualified stock-flow identity would have the form

$$Q_{t+1} = Q_t + I_t - O_t - X_t, \quad (19)$$

where Q_t is calls holding, I_t is inflow, O_t is ordinary outflow, and X_t is another qualified exit. Populating the identity requires calls-holding stocks, inflows, exits, and realized availability. Ordered priority histories and follow-up states add the remaining transitions.

Endpoint timestamps omit the stock waiting, the resources available, and the sequence of priority and exit changes. The product and queue examples show how each performance target selects its own denominator, clock, and witness requirements.

5.3 Operational and record-production friction

The observation map separates two forms of friction. Operational friction changes the service path through congestion, unit availability, assignment, handoffs, or follow-up. Record-production friction changes how those stages are classified, overwritten, retained, linked, or exported. Both can produce gaps in public timestamps or changes in public-state composition.

The Ida pattern is compatible with either source, or with both operating together. The diagnostic supplies no probability ordering over them. Target-specific witnesses localize the ambiguity: holding and availability histories recover operational congestion; joint public-internal counts recover capture; retained lineage reveals where records cease to share a denominator. A better release therefore improves measurement by contracting the relevant identified set. Whether a reform also reduces operational friction is a separate evaluative question.

6 From identified sets to admissible measures

Equation 13 turns five design choices into an admissibility test: labeling rule, denominator, operational clock, endpoint selection, and stage lineage. Equation 17 motivates lineage; Equation 19 motivates stocks and transitions. Ida makes these choices empirically consequential because the

shared public trace changed while its generating pathway remained set-valued.

Directly supportable administrative measures are functions of the released file under an explicit rule and denominator. Bounds propagate unresolved labels or missing outcomes. Selected measures describe the risk set on which their endpoints are observed. Measures requiring absent clocks, states, or links become available when the corresponding witnesses enter W .

Responsiveness requires a time origin, endpoint, and population at risk. Prioritization requires an ordered history when priorities change. Follow-through requires links from CAD to the applicable downstream stage, such as EPR production, program referral, or a policy-defined callback. A recorded disposition can mark an administrative branch, including GOA, while later duties require their own evidence.

NOPD policy anticipates assigned units, dispatch, arrival, return-to-service, and disposition in the communications record (New Orleans Police Department, 2021). Later oversight and court-directed reforms identify response time, calls holding, priority changes, officer availability, callbacks, accessibility, and data quality as operational objects (City of New Orleans and U.S. Department of Justice, 2024; U.S. District Court for the Eastern District of Louisiana, 2025; Office of the Independent Police Monitor, 2022, 2026). These documents establish why each object matters, while the public CAD files do not record callback eligibility or completion. Empirical measurement requires period-specific values and retained lineage.

Before applying the framework to reported DV-related calls, one further question is necessary: did later public-data enrichment supply the missing clocks and links, making the Ida lesson specific to the 2021 data regime?

7 Do newer public data close the measurement gap?

We retrieved and froze the official 2025 and 2026 Calls for Service datasets, their metadata and dictionaries, the current response-time dashboard model, current response policies, and the 2025 Electronic Police Report metadata (City of New Orleans, 2025, 2026; New Orleans Police Department, 2025, 2024, 2026). The comparison evaluates public measurement regimes.

The newer regime adds machine-queryable derived fields, including a candidate en-route duration and multiple dispositions. Current policies also clarify priority ordering, alternate response, policy-defined callbacks, and return-to-queue rules. The annual raw schema retains the 2021 structure: initial and recorded signal/priority endpoints, create/dispatch/arrival/closure timestamps, final disposition, and self-initiation.

The structural witnesses remain separate. Public documentation leaves the derived clocks, ordering, multiplicity, overwrite behavior, and historical continuity unresolved. Calls holding, realized unit availability, timestamped priority histories, applicable callback or fallback states, and public-internal dispatch lineage require internal aggregate evidence.

Table 4. Post-Ida public enrichment and the remaining structural boundary

Object	Public improvement by 2025–2026	Remaining boundary
Internal/public dispatch bridge	more public analytical surfaces and a 2025 CAD–EPR item link	qualified internal dispatch event and field-lineage map needed
Queue and availability	endpoint durations and response-time summaries	calls-holding stock, unit-status history, and realized availability needed
Priority	initial/recorded endpoints, change flags, and clearer current policy	ordered event history, actor, reason, and overwrite trail needed
Continuity and fallback	current policies describe alternate routes and policy-defined callbacks	realized uptime, fallback activation, and applicable callback-execution state needed

The current-data audit illustrates endpoint selection. The 2025 file contains 152,701 records with an arrival field but no dispatch field; the 2026 snapshot contains 97,573. A dispatch-to-arrival duration excludes these arrival-observed records. Dispatch-field completeness is approximately 41.8 percent in both years. Exact initial-versus-recorded priority disagreement is 38.7 percent in 2025 and 37.1 percent in the 2026 snapshot. These all-call values establish pervasive contemporary public-field missingness and endpoint change. Their scope is the observed audit periods.

The replay sharpens the Ida lesson. Public enrichment improves description; performance measurement still requires lineage, complete clocks, and common denominators. The DV application now carries those requirements through the reporting-to-follow-up chain.

8 Application to reported DV-related calls

The framework now turns to the substantive objective. Candidate DV measures begin with the public fields tested under Ida and extend to reports and follow-up when retained links permit. The application derives each measurement status from the evidence required at its stage.

8.1 The shared CAD layer and the reporting-to-follow-up chain

The project’s M7C selection map follows a reported DV-related call through priority, public dispatch and arrival fields, recorded disposition, Electronic Police Report (EPR) production, and possible DVU or AIR entry. The frozen machine-readable records are `M7C_DV_STAGE_MAP.json` and `M7C_FROZEN_SOURCE_INVENTORY.json`:

reported DV-related CAD → priority → public dispatch/arrival fields
→ recorded disposition → EPR → DVU/AIR.

Each arrow requires a retained link, and each downstream stage creates a new denominator. A DV label defines the first box. DVU audit compliance describes sampled DVU files; AIR describes an advocacy-eligible pathway after program launch. Upstream lineage is needed to express either stage as a share of reported DV-related CAD records.

8.2 Reported demand begins at the CAD denominator

The workload estimand begins with calls created in CAD, D_t^{CAD} . It therefore measures reported demand recorded by the dispatch system. Help-seeking attempts that never reach the system,

contacts received without a retained CAD record, and underlying DV need occupy earlier stages of the demand chain. Measuring those stages requires external access, survey, hotline, or validation evidence.

This distinction also governs interpretation of call volume. A change in released counts can reflect help-seeking, access, CAD creation, classification, retention, or the underlying phenomenon. R12.2 estimates the system-level Ida record-state pattern; it reports no DV call-count trend. The DV application begins at the declared CAD denominator and asks how far performance measurement can proceed from there.

8.3 Classification defines the first denominator

Classification defines the first denominator. The available map lacks a fully qualified, period-specific frozen signal list. A declared rule therefore places records into three groups: DV-related, other, and unresolved. The unresolved group enters the bounds directly.

Let $\mathcal{L}$ denote the declared labeling rule and let $C_i^{\mathcal{L}} \in \{1, 0, U\}$ denote its three states: labeled DV-related, labeled not DV-related, and unresolved. If $N_1 = \sum_i \mathbf{1}\{C_i^{\mathcal{L}} = 1\}$ and $N_U = \sum_i \mathbf{1}\{C_i^{\mathcal{L}} = U\}$, the sharp count bound is

$$N_{DV}^{\mathcal{L}} \in [N_1, N_1 + N_U]. \quad (20)$$

For a fixed CAD denominator N , the corresponding share lies in $[N_1/N, (N_1 + N_U)/N]$. The superscript ties the estimand to rule $\mathcal{L}$. Appendix Section 6.8 gives the labeling semantics and sharpness argument.

At the disposition stage, GOA is a recorded administrative branch. It becomes performance-relevant when policy attaches a specific downstream duty, such as report production or an applicable callback; completion of that duty supplies the corresponding outcome.

8.4 Response-time selection and sharp full-denominator bounds

Let $E_i = 1$ when the timestamps required for a declared response-time clock are observed. Among records labeled DV-related under $\mathcal{L}$, define

$$\pi_E = \Pr(E = 1 \mid C^{\mathcal{L}} = 1), \quad F_E(a) = \Pr(\tau \leq a \mid C^{\mathcal{L}} = 1, E = 1). \quad (21)$$

The public file directly supports the selected distribution F_E . The desired distribution over the full declared-rule denominator is

$$F_{\tau}(a) = \pi_E F_E(a) + (1 - \pi_E) F_0(a), \quad (22)$$

where F_0 is the unobserved distribution for endpoint-missing records. Allowing $F_0(a)$ to range over $[0, 1]$ gives the sharp pointwise bounds

$$\pi_E F_E(a) \leq F_{\tau}(a) \leq \pi_E F_E(a) + (1 - \pi_E). \quad (23)$$

These bounds state exactly what endpoint selection costs. Qualified endpoint recovery or a defensible restriction on the missing group narrows the interval. Unresolved U labels enlarge the outer set through equation (20).

8.5 Priority histories and administrative follow-through

Initial and recorded priorities reveal endpoint disagreement. If $H_i = 1$ denotes at least one priority change, then

$$\Pr(P^0 \neq P^r) \leq \Pr(H = 1) \leq 1. \quad (24)$$

Ordered priority histories replace this bound with measured transitions, their timing, and their direction.

For follow-through, let $L_i = 1$ indicate retained lineage from the CAD denominator to the declared follow-up stage and let $Z_i = 1$ indicate completion of the specified administrative duty. Define

$$\pi_L = \Pr(L = 1 \mid C^{\mathcal{L}} = 1), \quad p_L = \Pr(Z = 1 \mid C^{\mathcal{L}} = 1, L = 1). \quad (25)$$

The full-denominator completion rate p_Z obeys

$$\pi_L p_L \leq p_Z \leq \pi_L p_L + (1 - \pi_L). \quad (26)$$

When lineage coverage is unavailable, the public-data set is $[0, 1]$. Aggregate retained lineage identifies π_L ; linked completion counts identify p_L and close the remaining gap.

8.6 What current evidence can support

Table 5. Measurement frontier along the reported-DV administrative chain

Class	Candidate measure	Supported interpretation
Directly supportable administrative measure	Rule-defined reported-DV counts and $1/0/U$ bounds; recorded disposition composition; public-field completeness	Functions of released fields on an explicit CAD denominator. The declared rule and unresolved class remain part of the estimand.
Bounded or selected measure	Response times among records with the required endpoints; initial and recorded priority endpoints; CAD–EPR key presence; DVU or AIR measures on their own denominators	Equations 23–26 propagate endpoint and lineage coverage; downstream programs retain their selected denominators.
Unavailable from public CAD	Physical service stages; queue delay; realized unit availability or effective capacity; policy-defined callback completion; response quality; victim safety; true DV incidence; a unique mechanism	Internal states, retained links, or separate outcome evidence are required to define these targets.

The table applies [equation \(13\)](#). Classification uncertainty produces count bounds, endpoint missingness produces selected distributions and full-denominator bounds, and downstream programs define new populations. Retained lineage places these components on a common reported-call denominator.

8.7 Target-specific privacy-safe witness sets

Minimality is target-specific because each performance measure uses a different denominator, clock, or completion rule. Appendix Section 1.3 formalizes the relevant standard. Building on observational equivalence and partial identification, a witness bundle is sufficient for a declared target when it defines the risk set and contracts the target’s identified set to the prespecified precision. It is inclusion-minimal when removing any member restores excess ambiguity or leaves the target undefined. We call this a target-specific witness sufficiency and necessity analysis. Minimum cost or minimum disclosure are separate implementation criteria and can select among several inclusion-minimal bundles.

Qualified period-specific DV labeling and retained event lineage form the common core. The lineage remains internal: its role is to produce reconciled aggregate cells on the same reported-call denominator. The measure-specific additions are shown in Table 6.

Table 6. Target-specific privacy-safe witness bundles

Target	Required joint witness	Additional information for a broader target
Reported-DV risk set	Qualified period-specific labeling rule, pre-specified treatment of U , and internal lineage capable of producing reconciled aggregate cells	Separate validation evidence if the target changes from a declared administrative label to underlying DV status
Receipt-to-assignment delay	CAD receipt and first qualified assignment times for the same calls, with endpoint coverage on the full denominator	Realized unit availability for capacity-adjusted delay or an operational explanation
Receipt-to-qualified-arrival responsiveness	CAD receipt and qualified operational arrival endpoints for the same calls, with endpoint coverage	Calls-holding and availability histories when the target decomposes total time into queue and travel components
Priority revision	Initial priority and the complete ordered priority history, including event times and direction	Actor and reason fields for process or rule-compliance analysis
Administrative follow-through	Duty eligibility, retained CAD-stage lineage, and completion status or time for the applicable downstream stage (EPR, program referral, or policy-defined callback)	Content or quality evidence when the target extends beyond completion of the declared duty

The required evidence is joint rather than marginal. A privacy-safe release must preserve cells such as reported-DV calls by endpoint coverage and duration bin, or eligible calls by lineage and completion status. Public event identifiers are unnecessary when the agency constructs those cells internally and applies small-cell suppression.

The necessity arguments clarify the boundary. Removing the labeling rule changes the reported-DV risk set. Removing lineage permits the same marginal clocks or completion totals to be matched to different CAD calls. Removing one response clock leaves duration undefined; incomplete endpoint coverage leaves a selected distribution and the bounds in [equation \(23\)](#). Endpoint priorities permit offsetting changes, whereas an ordered history measures them. Follow-through requires both duty

eligibility and completion on the common denominator.

Calls-holding transitions therefore belong to the minimum bundle for queue delay, while realized unit availability belongs to a capacity-adjusted target. A fully observed receipt-to-qualified-endpoint distribution uses neither. These target-complete bundles support performance measurement conditional on reported DV-related calls received by CAD. Population incidence, victim safety, and mechanism remain distinct estimands.

8.8 A DV-specific estimator would be a new empirical extension

A DV-subset version of the ten-bin event–reference estimator is a separate empirical extension. It requires qualified period-specific classification authority, prespecified treatment of U records, support and cell-size floors, prespecified reference eligibility, and selection handling for endpoints and downstream stages. Its reference set would be requalified for the DV estimand. R12.2 reports only the frozen system-level estimates.

9 Implications for resource allocation and civilian oversight

The witness analysis turns the measurement frontier into a release design. The same aggregate evidence serves two users. Police managers need common-denominator flows to align capacity with reported demand and locate delay. Civilian oversight needs the same flows to distinguish congestion, prioritization, endpoint coverage, and administrative follow-through.

Table 7. A shared measurement layer for management and oversight

Aggregate object	Resource-management use	Civilian-oversight use
Reported calls by qualified label and priority	Describe incoming workload and priority mix	Reconcile the population used in published measures
Calls holding and realized availability over time	Locate queue pressure and capacity constraints	Separate waiting from endpoint or recording gaps
Receipt, assignment, and qualified endpoint distributions	Compare delay across declared stages and risk sets	Audit endpoint coverage and selected response summaries
CAD-applicable-stage lineage and completion cells	Locate administrative attrition and follow-up workload	Verify completion rates on the original eligible-call denominator

These objects can be released as time-binned joint counts, coverage rates, transition matrices, and duration distributions with small-cell suppression. The design requires neither victim identifiers, unit identifiers, exact locations, nor live deployment information. Stable definitions, versioned labeling rules, and reconciled denominators allow managers and reviewers to reproduce each published measure.

The payoff has two stages. Better releases contract identified sets immediately, making descriptive performance measures more precise and comparable. Richer lineage can then support evaluation

of staffing, dispatch, reporting, or policy-defined follow-up reforms when paired with a credible experimental or quasi-experimental design. Ida is unsuitable as an instrument for those effects because it jointly perturbed demand, operations, and recording. Its role is to reveal the measurement problem that a later causal design must first solve.

10 Limitations

The Ida statistic measures rarity within frozen comparison sets; the Kitagawa calculation is an accounting decomposition, and the institutional timeline supplies context. The event jointly stressed operations and record production, so the generating pathway remains set-valued. The 2025–2026 audit describes its observed public-data regime. Help-seeking, access, evacuation, mobility, holidays, weather, COVID-era conditions, redaction, and suppression can also change reported-call populations.

The DV application establishes a measurement frontier. Exact period-specific signal-list authority remains unresolved where noted, and downstream sources condition on selected populations. An empirical DV performance estimate requires qualified classification and the aggregate lineage specified above.

11 Conclusion

Police workload theory links reported demand, effective capacity, priorities, and service. Ida establishes that the public observation of that system can change in an extreme, structured, within-disposition way. Its score is reference-extreme, while the service and record-production paths that generated the pattern remain observationally equivalent.

The measure-specific identified set carries that finding to reported DV-related calls. Declared-rule administrative counts and $1/0/U$ bounds are directly supportable. Endpoint-observed response times describe a selected risk set, with sharp full-denominator bounds determined by coverage. Ordered histories recover priority revision. Retained CAD-to-applicable-downstream-stage lineage places administrative follow-through on the original eligible-call denominator.

Qualified labeling and privacy-safe aggregate lineage form the common measurement core. Identified-set contraction and witness-necessity arguments then select the clocks, histories, eligibility rules, completion fields, holding stocks, and availability measures required by particular estimands. Released as suppressed aggregate joint cells, these witnesses let managers locate workload pressure and let civilian reviewers audit the same denominators and transitions.

The motivating question therefore has a conditional answer. Public CAD supports defensible administrative measures whose labels, fields, and risk sets are explicit; responsiveness, prioritization, and follow-through require the additional witnesses derived here. Those data would support an empirical DV performance study and stronger future evaluations of resource allocation. True DV incidence, victim safety, and the mechanism behind the Ida pattern remain separate targets.

References

- Adams, A., Huttunen, K., Nix, E., and Zhang, N. (2024). The dynamics of abusive relationships. *The Quarterly Journal of Economics*, 139(4):2135–2180.
- Ahmed, H., Latif, D., and Vyborny, K. (2026). The unintended consequences of accountability: Quasi-experimental evidence from policing in pakistan. *Journal of Public Economics*, 258:105614.

- Alvarez, F. E., Argente, D., and Van Patten, D. (2026). On the resilience of payment methods. Working Paper 35115, National Bureau of Economic Research.
- Beland, L.-P. (2026). Police workload, responsiveness, and public-safety outcomes. Unpublished conceptual presentation.
- Blackwell, D. (1953). Equivalent comparisons of experiments. *Annals of Mathematical Statistics*, 24(2):265–272.
- Blanes i Vidal, J. and Kirchmaier, T. (2018). The effect of police response time on crime clearance rates. *The Review of Economic Studies*, 85(2):855–891.
- Boivin, R. and Cordeau, G. (2011). Measuring the impact of police discretion on official crime statistics: A research note. *Police Quarterly*, 14(2):186–203.
- Boivin, R. and Ouellet, F. (2014). Space and time variations in crime-recording practices within a large municipal police agency. *International Journal of Police Science & Management*, 16(3):171–183.
- Bullinger, L. R., Carr, J. B., and Packham, A. (2021). COVID-19 and crime: Effects of stay-at-home orders on domestic violence. *American Journal of Health Economics*, 7(3):249–280.
- Carvalho, V. M., Nirei, M., Saito, Y. U., and Tahbaz-Salehi, A. (2021). Supply chain disruptions: Evidence from the great east japan earthquake. *Quarterly Journal of Economics*, 136(2):1255–1321.
- Chen, X., Hong, H., and Nekipelov, D. (2011). Nonlinear models of measurement errors. *Journal of Economic Literature*, 49(4):901–937.
- City of New Orleans (2021). Calls for service 2021. Data.NOLA dataset 3pha-hum9.
- City of New Orleans (2025). Calls for service 2025. Data.NOLA dataset 4xwx-sfte.
- City of New Orleans (2026). Calls for service 2026. Data.NOLA dataset es9j-6y5d.
- City of New Orleans and U.S. Department of Justice (2024). New orleans police department sustainment plan. Filed September 27, 2024.
- Courty, P. and Marschke, G. (1997). Measuring government performance: Lessons from a federal job-training program. *American Economic Review*, 87(2):383–388.
- DeAngelo, G., Toger, M., and Weisburd, S. (2023). Police response time and injury outcomes. *The Economic Journal*, 133(654):2147–2177.
- Dominitz, J. and Sherman, R. P. (2006). Identification and estimation of bounds on school performance measures: A nonparametric analysis of a mixture model with verification. *Journal of Applied Econometrics*, 21(8):1295–1326.
- Eckhouse, L. (2022). Metrics management and bureaucratic accountability: Evidence from policing. *American Journal of Political Science*, 66(2):385–401.

- European Institute for Gender Equality (2023). Methodological guidance: Administrative data collection on violence against women and domestic violence. Technical report, Publications Office of the European Union.
- Horowitz, J. L. and Manski, C. F. (1995). Identification and robustness with contaminated and corrupted data. *Econometrica*, 63(2):281–302.
- Kapteyn, A. and Ypma, J. Y. (2007). Measurement error and misclassification: A comparison of survey and administrative data. *Journal of Labor Economics*, 25(3):513–551.
- Klinger, D. A. and Bridges, G. S. (1997). Measurement error in calls-for-service as an indicator of crime. *Criminology*, 35(4):705–726.
- Knox, D., Lowe, W., and Mummolo, J. (2020). Administrative records mask racially biased policing. *American Political Science Review*, 114(3):619–637.
- Koopmans, T. C. and Reiersøl, O. (1950). The identification of structural characteristics. *Annals of Mathematical Statistics*, 21(2):165–181.
- Leslie, E. and Wilson, R. (2020). Sheltering in place and domestic violence: Evidence from calls for service during covid-19. *Journal of Public Economics*, 189:104241.
- Lewbel, A. (2019). The identification zoo: Meanings of identification in econometrics. *Journal of Economic Literature*, 57(4):835–903.
- Manski, C. F. (1989). Anatomy of the selection problem. *Journal of Human Resources*, 24(3):343–360.
- Manski, C. F. (1990). Nonparametric bounds on treatment effects. *American Economic Review*, 80(2):319–323.
- Manski, C. F. (2003). *Partial Identification of Probability Distributions*. Springer, New York.
- Manski, C. F. (2015). Communicating uncertainty in official economic statistics: An appraisal fifty years after morgenstern. *Journal of Economic Literature*, 53(3):631–653.
- Masten, M. A. and Poirier, A. (2020). Inference on breakdown frontiers. *Quantitative Economics*, 11(1):41–111.
- Miller, A. R. and Segal, C. (2019). Do female officers improve law enforcement quality? effects on crime reporting and domestic violence. *The Review of Economic Studies*, 86(5):2220–2247.
- Miller, A. R., Segal, C., and Spencer, M. K. (2022). Effects of COVID-19 shutdowns on domestic violence in US cities. *Journal of Urban Economics*, 131:103476.
- Miller, A. R., Segal, C., and Spencer, M. K. (2024). Effects of the COVID-19 pandemic on domestic violence in los angeles. *Economica*, 91(361):163–187.
- Molinari, F. (2008). Partial identification of probability distributions with misclassified data. *Journal of Econometrics*, 144(1):81–117.

- Molinari, F. (2020). Microeconometrics with partial identification. In *Handbook of Econometrics*, volume 7A, pages 355–486. Elsevier.
- Moore, M. H. and Braga, A. A. (2004). Police performance measurement: A normative framework. *Criminal Justice Ethics*, 23(1):3–19.
- Mourtgos, S. M., Adams, I. T., and Nix, J. (2024). Staffing levels are the most important factor influencing police response times. *Policing: A Journal of Policy and Practice*, 18:paae002.
- New Orleans Police Department (2021). Regulations manual. Chapters 41.4.1, 81.1, 81.4, 82.1.1, and 82.4.
- New Orleans Police Department (2024). Chapter 41.4.2: Alternative police response. Effective September 15, 2024.
- New Orleans Police Department (2025). Chapter 41.4.1: Response to police calls. Revised policy.
- New Orleans Police Department (2026). Calls for service response times dashboard. Public Power BI report, model refreshed July 17, 2026.
- Office of the Independent Police Monitor (2021). 2021 hurricane season police oversight report: Understanding hurricane ida and policing response. City of New Orleans.
- Office of the Independent Police Monitor (2022). 2022 hurricane and emergency oversight plan. City of New Orleans.
- Office of the Independent Police Monitor (2026). 2026 hurricane and emergency oversight plan. City of New Orleans.
- Pandey, S. (2025). Coding bias: The role of racial-ethnic identity in 911 call dispatching decisions. Working paper, The Ohio State University.
- Peters, J., Bühlmann, P., and Meinshausen, N. (2016). Causal inference by using invariant prediction: Identification and confidence intervals. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 78(5):947–1012.
- Phoenix, J. (2023). Improving police data collection to measure repeat demand: A focus on domestic violence and abuse. *Policing: A Journal of Policy and Practice*, 17:paad022.
- Rollings, K. and Taylor, N. (2008). Measuring police performance in domestic and family violence. Technical Report 367, Australian Institute of Criminology.
- Rothenberg, T. J. (1971). Identification in parametric models. *Econometrica*, 39(3):577–591.
- Simpson, R. and Orosco, C. (2021). Re-assessing measurement error in police calls for service: Classifications of events by dispatchers and officers. *PLOS ONE*, 16(12):e0260365.
- Tamer, E. (2010). Partial identification in econometrics. *Annual Review of Economics*, 2:167–195.
- Tiwana, N., Bass, G., and Farrell, G. (2015). Police performance measurement: An annotated bibliography. *Crime Science*, 4(1):1.

- U.S. District Court for the Eastern District of Louisiana (2025). Order and reasons entering sustainment plan, united states v. city of new orleans, no. 12-1924. Document 822, filed January 14, 2025.
- Verlaan, T. and Ruiter, S. (2023). Predictors of police response time: A scoping review. *Crime Science*, 12:19.
- Vomfell, L. and Povala, J. (2023). (no) spillovers in reporting domestic abuse to police. *Journal of the Royal Statistical Society Series A: Statistics in Society*, 186(4):811–833.

Mathematical Definitions, Identification Results, and Reproduction Guide

Appendix to *From Police Workload to Auditable Performance Measures*

24 August 2026

Appendix R12.2 Results R12

Purpose and reading guide

This appendix connects a police-workload layer to two mathematical modules. Section 1 defines the reported-demand denominator, the public observation map, measure-specific identified sets, and target-specific witness minimality. Sections 2–4 define the public-record stress-test statistic, and Section 5 localizes the Ida pattern. Section 6 applies the identified-set logic to the internal/public bridge, queue, priority, DV classification, endpoint-selected response time, and follow-through. Section 7 retains the prospective multi-model frontier. Section 8 formalizes the 2025–2026 measurement-regime replay. Sections 10–11 record numerical invariants, reproduction commands, and interpretation rules. Every reported empirical value comes from the locked aggregate artifacts.

Contents

1	Measurement targets, released records, and identified sets	2
2	Observable universe, clocks, and public record states	5
3	Topology matrix and constrained discrepancy score	7
4	Reference-window construction and empirical rarity	9
5	Within-disposition localization and exact accounting	11
6	Measure-specific structural partial identification	14
7	Prospective technical frontier: model uncertainty and reachability	18
8	Current-public-data measurement-regime replay	21
9	Statistical and econometric interpretation	23
10	Reproduction guide	24
11	Claim language and interpretation rules	28
12	Summary of the mathematical logic	28

1 Measurement targets, released records, and identified sets

1.1 Observation map and candidate performance functionals

Let S denote a latent operational and administrative-follow-through path for the reported-call population of interest. Let R denote the recording, retention, and export process. The released public record is

$$Y = h(S, R). \quad (1)$$

For candidate measure k , let

$$\theta_k = m_k(S) \quad (2)$$

be the corresponding performance functional. Examples include a response-time distribution on a declared call denominator, the prevalence of priority changes, and completion of a specified follow-through duty.

Let W collect the admitted semantic and operational witnesses. These witnesses restrict the feasible pairs (S, R) to a compatibility set $\mathcal{C}(Y, W)$. The measure-specific identified set is

$$\mathcal{I}_k(Y, W) := \{m_k(S) : (S, R) \in \mathcal{C}(Y, W)\}. \quad (3)$$

Measure-specific identification rule

The public record directly supports θ_k when $\mathcal{I}_k(Y, W)$ is a singleton. It partially identifies θ_k when the set is informative but contains multiple values. An observed statistic is selected when its risk set is restricted by endpoint or stage presence. If the target stage or denominator cannot be formed under W , the measure is unavailable from the current record.

If W' contains stronger witnesses than W , then $\mathcal{C}(Y, W') \subseteq \mathcal{C}(Y, W)$ and therefore

$$\mathcal{I}_k(Y, W') \subseteq \mathcal{I}_k(Y, W). \quad (4)$$

Thus retained lineage, joint counts, complete clocks, and ordered histories have a precise role: they remove observationally equivalent paths. The construction begins with the classical definition of identification through observational equivalence (Koopmans and Reiersøl, 1950; Rothenberg, 1971; Lewbel, 2019) and follows partial-identification analysis by retaining every target value compatible with the data and maintained restrictions (Manski, 1989, 1990, 2003; Molinari, 2020). Its administrative-data interpretation follows work that distinguishes verified from unverified records and propagates that distinction into performance bounds (Dominitz and Sherman, 2006; Kapteyn and Ypma, 2007; Manski, 2015).

1.2 Workload layer and two-sided observability

The operational path begins with the demand recorded by CAD. Let D_t^* denote latent need, H_t^{seek} help-seeking attempts, A_t^{recv} contacts received by the emergency system, and D_t^{CAD} calls created in CAD. The demand observation chain is

$$D_t^* \longrightarrow H_t^{\text{seek}} \longrightarrow A_t^{\text{recv}} \longrightarrow D_t^{\text{CAD}}. \quad (5)$$

Each arrow can select or transform the preceding population. Unless a target states otherwise, the paper conditions on D_t^{CAD} and leaves the earlier populations outside its denominator.

Let C_t denote effective operational capacity and P_t the priority process. A workload representation is

$$S_t = g(D_t^{\text{CAD}}, C_t, P_t), \quad (6)$$

followed by the public observation map $Y_t = h(S_t, R_t)$. Equation (6) organizes service; Equation (1) determines whether the released record measures it. A performance estimand can therefore be written explicitly as

$$\theta_k = m_k(S \mid D^{\text{CAD}}), \quad (7)$$

with the clock, endpoint, and eligible population declared for target k .

Two kinds of friction enter this system. Operational friction changes S through holding, availability, assignment, handoffs, or follow-up. Record-production friction changes R through classification, overwrite, retention, linkage, or export. Distinct pairs (S, R) can generate the same Y , so public missingness alone cannot allocate the discrepancy between these sources. Holding and availability histories, joint public-internal counts, and retained stage lineage contract the relevant alternatives through [equation \(4\)](#).

The released call count is likewise a measure of D_t^{CAD} , rather than a direct measure of D_t^* or either intermediate stage in [equation \(5\)](#). R12.2 estimates no DV call-volume trend. Its DV application begins with a declared CAD risk set and derives the additional witnesses needed to measure service conditional on that set.

1.3 Target-specific sufficiency and inclusion-minimality

Fix a target k , a metric ρ_k on its parameter space, and a prespecified tolerance $\delta_k \geq 0$. For any witness bundle W , define

$$\text{diam}_{\rho_k} \mathcal{I}_k(Y, W) := \sup_{a, b \in \mathcal{I}_k(Y, W)} \rho_k(a, b). \quad (8)$$

A bundle is sufficient for target k when the target population and stage are defined under W and

$$\text{diam}_{\rho_k} \mathcal{I}_k(Y, W) \leq \delta_k. \quad (9)$$

The choice $\delta_k = 0$ requires point identification. A positive tolerance permits a declared partial-identification standard.

A sufficient bundle W_k^* is inclusion-minimal when

$$W_k^* \setminus \{w\} \text{ is insufficient for every } w \in W_k^*. \quad (10)$$

Equivalently, witness $w \in W$ is necessary relative to bundle W and tolerance δ_k when removing it leaves the target undefined or

$$\text{diam}_{\rho_k} \mathcal{I}_k(Y, W \setminus \{w\}) > \delta_k. \quad (11)$$

The argument is constructive: necessity is established by exhibiting two paths that agree on Y and the retained witnesses but produce target values farther apart than δ_k .

This formulation connects information comparison to sensitivity analysis. Additional signals refine an experiment in the sense developed by [Blackwell \(1953\)](#); relaxing assumptions until a conclusion

fails motivates the breakdown-frontier approach of Masten and Poirier (2020). The present analysis applies these ideas to a finite bundle of administrative observations and a declared performance target. We use *witness sufficiency and necessity analysis* to distinguish this identification argument from statistical hypothesis testing.

Several inclusion-minimal bundles may coexist. If $c(w)$ records production, privacy, or disclosure burden and $\mathfrak{P}$ is the admissible privacy class, a minimum-cost design instead solves

$$\begin{aligned} \min_{W \subseteq \mathbb{W}} \quad & \sum_{w \in W} c(w) \\ \text{subject to} \quad & W \in \mathfrak{P}, \quad k \text{ is defined under } W, \\ & \text{diam}_{\rho_k} \mathcal{I}_k(Y, W) \leq \delta_k. \end{aligned} \tag{12}$$

The paper establishes target-specific inclusion-minimality; an agency can apply equation (12) using its own production and privacy costs.

Let w_C denote a qualified period-specific DV labeling rule with prespecified treatment of U ; w_L internal event lineage capable of producing reconciled aggregate cells; w_0 the declared response-clock origin; w_1 its qualified endpoint; w_E endpoint coverage on the full denominator; w_P the complete ordered priority history; w_A follow-through eligibility; and w_Z completion status or time. Under no auxiliary restrictions, the following necessity results hold.

Target-specific witness-necessity results

For response time, $\{w_C, w_L, w_0, w_1\}$ identifies the endpoint-observed distribution on its selected risk set; adding w_E identifies the coverage share and therefore the sharp full-denominator bounds. Point identification of the full-denominator distribution additionally requires complete endpoints or a separately justified restriction on missing durations. Deleting w_0 or w_1 leaves duration undefined, while deleting w_L permits the same marginal clock distributions to be paired into different duration distributions.

For the prevalence of any priority revision, $\{w_C, w_L, w_P\}$ is inclusion-minimal. With only initial and recorded endpoints, an equal-endpoint call may either remain unchanged or follow an offsetting sequence and return to its initial value.

For an administrative follow-through rate, $\{w_C, w_L, w_A, w_Z\}$ is inclusion-minimal. Deleting w_A changes the eligible denominator; deleting w_L allows the same completion total to be reassigned across CAD calls; deleting w_Z leaves completion undefined.

These are target-specific results. Calls-holding transitions enter the minimum bundle when the target is queue delay. Realized unit availability enters when the target includes capacity or capacity-adjusted responsiveness. A fully observed receipt-to-qualified-endpoint distribution uses neither. Wherever w_L appears, the privacy-safe product must retain the indicated joint cells; marginal aggregates alone leave the pairing unresolved, although no event identifier need be released.

1.4 The stress-test statistic is an observable diagnostic

The Ida statistic belongs to the released-record side of [equation \(1\)](#). Let $G(Y)$ be the standardized ten-bin public-topology matrix defined below. Then

$$T(Y) := U_{\text{full}}(G(Y)) \quad (13)$$

is a constrained discrepancy statistic applied identically to Ida and every qualified reference window. It measures whether the released field relationships remain close to the frozen support class. It does not replace $m_k(S)$.

This separation fixes the inferential order. Ida establishes that the observed product Y can become reference-extreme. [Equation \(3\)](#) determines what that product reveals about each candidate performance functional. Disaster-based studies often use a shock to expose system dependencies, but stronger operational conclusions come from linked outcomes, infrastructure measures, or structural restrictions ([Carvalho et al., 2021](#); [Alvarez et al., 2026](#)). Invariance across environments can carry causal content under additional assumptions; the present statistic uses environment comparison only to diagnose stability of released-record relationships ([Peters et al., 2016](#)).

2 Observable universe, clocks, and public record states

2.1 Units and indices

Let i index a public calls-for-service record. Let $z \in \{\text{E}, \text{R}\}$ index the Ida event window and a qualified reference window. Let s index the frozen common-support standardization strata, $t \in \{1, \dots, 10\}$ index the ten twelve-hour bins, and r index a recorded disposition category.

The event window is

$$[2021-08-29 \text{ 00:00}, 2021-09-03 \text{ 00:00}) \quad (14)$$

in local Central Daylight Time. The fixed bins are listed in [table 1](#).

Table 1. Complete calendar map for the ten frozen twelve-hour bins

Bin	Start	End
B1	29 Aug. 2021, 00:00	29 Aug. 2021, 12:00
B2	29 Aug. 2021, 12:00	30 Aug. 2021, 00:00
B3	30 Aug. 2021, 00:00	30 Aug. 2021, 12:00
B4	30 Aug. 2021, 12:00	31 Aug. 2021, 00:00
B5	31 Aug. 2021, 00:00	31 Aug. 2021, 12:00
B6	31 Aug. 2021, 12:00	1 Sep. 2021, 00:00
B7	1 Sep. 2021, 00:00	1 Sep. 2021, 12:00
B8	1 Sep. 2021, 12:00	2 Sep. 2021, 00:00
B9	2 Sep. 2021, 00:00	2 Sep. 2021, 12:00
B10	2 Sep. 2021, 12:00	3 Sep. 2021, 00:00

Identical relative-position bins are used for each pseudo-event reference window. The bin definitions were frozen before the focal cell analysis.

2.2 Public field-presence indicators

For each public record i , define

$$D_i = \mathbf{1}\{\text{the public dispatch field passes the frozen validity rule}\}, \quad (15)$$

$$A_i = \mathbf{1}\{\text{the public arrival field passes the frozen validity rule}\}. \quad (16)$$

These are Type-F public measurement functionals: they are deterministic functions of the qualified public fields. They are not latent operational events.

For $d, a \in \{0, 1\}$, define the record state

$$J_{da,i} = \mathbf{1}\{D_i = d, A_i = a\}. \quad (17)$$

The four indicators satisfy the recordwise simplex identity

$$J_{00,i} + J_{01,i} + J_{10,i} + J_{11,i} = 1. \quad (18)$$

The public arrival and dispatch indicators can be recovered by

$$A_i = J_{01,i} + J_{11,i}, \quad D_i = J_{10,i} + J_{11,i}. \quad (19)$$

Interpretive firewall

$J_{01,i} = 1$ means that the public arrival field is present and the public dispatch field is absent under the frozen parsing rules. It does not assert that a unit physically arrived without being dispatched. Likewise, $J_{11,i} = 1$ does not prove that a complete physical sequence occurred.

2.3 Risk sets, shares, and common-support standardization

Let $\mathcal{R}_{st}^z$ be the qualified record set in population z , stratum s , and bin t . The unstandardized state share is

$$p_{jst}^z = \frac{\sum_{i \in \mathcal{R}_{st}^z} J_{j,i}}{|\mathcal{R}_{st}^z|}, \quad j \in \{00, 01, 10, 11\}. \quad (20)$$

Let $\mathcal{S}_t$ be the frozen bin-specific common-support stratum set and ω_{st}^R the reference-derived weights, normalized within bin:

$$\omega_{st}^R \geq 0, \quad \sum_{s \in \mathcal{S}_t} \omega_{st}^R = 1. \quad (21)$$

The standardized share is

$$\bar{p}_{j,t}^z = \sum_{s \in \mathcal{S}_t} \omega_{st}^R p_{jst}^z, \quad (22)$$

and the event-reference contrast is

$$\Delta_{j,t} = \bar{p}_{j,t}^E - \bar{p}_{j,t}^R = \sum_{s \in \mathcal{S}_t} \omega_{st}^R (p_{jst}^E - p_{jst}^R). \quad (23)$$

Because the shares form a simplex in each stratum and population,

$$\sum_{j \in \{00,01,10,11\}} \Delta_{j,t} = 0 \quad \text{for every } t. \quad (24)$$

The standardized public marginals obey

$$\Delta_{A,t} = \Delta_{01,t} + \Delta_{11,t}, \quad (25)$$

$$\Delta_{D,t} = \Delta_{10,t} + \Delta_{11,t}. \quad (26)$$

These identities are numerical invariants in every valid reconstruction.

3 Topology matrix and constrained discrepancy score

3.1 The ten-bin matrix

Because $\Delta_{00,t}$ is determined by equation (24), the primary matrix retains three independent coordinates:

$$G = \begin{bmatrix} \Delta_{01,1} & \Delta_{10,1} & \Delta_{11,1} \\ \vdots & \vdots & \vdots \\ \Delta_{01,10} & \Delta_{10,10} & \Delta_{11,10} \end{bmatrix} \in \mathbb{R}^{10 \times 3}. \quad (27)$$

Let g_j denote column j . Independent reconstruction of the locked Ida matrix achieved maximum absolute parity error 4.13×10^{-13} and simplex error 1.11×10^{-16} .

3.2 Authorized support triples

Let $\mathcal{X}$ be the finite set of authorized support triples. Each

$$X = [x_1, x_2, x_3] \in \mathbb{R}^{10 \times 3} \quad (28)$$

encodes a prespecified temporal support representation.

3.2.1 Entry construction and frozen finite library

For the full-combined class, the three columns are ordered as twelve-hour shifts, redeployment/anti-looting, and curfew. Let e_k denote an authorized chronology state for family k , and let

$$I_{e_k, \lambda} = [a_{e_k}, b_{e_k} + \lambda) \quad (29)$$

be its half-open support interval after end-only propagation by

$$\lambda \in \{0, 6, 12, 18, 24, 36, 48, 60, 72\} \text{ hours}. \quad (30)$$

Write $N_{st}^E = |\mathcal{R}_{st}^E|$ and let $N_{st}^{E, \text{in}}(e_k, \lambda)$ count records in $\mathcal{R}_{st}^E$ whose public creation time lies in $I_{e_k, \lambda}$. The matrix entry is

$$X_{tk} = \omega_t(e_k, \lambda) = \sum_{s \in \mathcal{S}_t} \omega_{st}^R \frac{N_{st}^{E, \text{in}}(e_k, \lambda)}{N_{st}^E}. \quad (31)$$

Thus each entry is the reference-standardized share of the bin's event records temporally covered by a frozen institutional-event or operational-regime interval. It is a temporal-support coverage share,

not a causal exposure, service level, or capacity measure.

The chronology adjudication is recorded in `WAVE4R_CHRONOLOGY_SETS.json`. Relative to the Wave4R experiment root, the materialized values and deduplicated state counts are recorded in `candidate/r4_replication/child_e4_run/R4_CHILD_E4_SUPPORT.csv` and `candidate/r4_replication/child_e4_run/R4_CHILD_E4_SUPPORT_STATE_REGISTRY.json`. Because first-hand documentation identifies the shifts, redeployment, and curfew starts only at the date level, each is represented by the exact finite family induced by eligible event-record times and the boundaries of that date; states with the same full ten-bin signature are then deduplicated. This retains the documented temporal ambiguity rather than inventing an unsupported exact start time.

At each frozen horizon, the three operational-regime families contain 864, 610, and 632 unique ten-bin signatures. Their Cartesian product therefore contains

$$864 \times 610 \times 632 = 333,089,280 \quad (32)$$

authorized triples. The nine horizon-specific support-vector sets are invariant, so M7A uses $\lambda = 0$ as the representative horizon. The file `candidate/m7a_full_combined_unified_threshold/M7A_TRIPLE_COVERAGE_CERTIFICATE.json` certifies complete branch-and-bound coverage of this product for the 0.25 exclusion decision. Complete coverage means that every authorized triple is represented by the certified partition and pruning argument; it does not mean that a separate leaf linear program was solved for every triple.

3.2.2 Rowwise envelope

For row t , define the support envelope

$$\omega_t(X) = \max_{k \in \{1,2,3\}} X_{tk}. \quad (33)$$

The envelope is the rowwise maximum of the three reference-standardized temporal-support coverage shares. It is a mathematical support-capacity object, not police capacity.

3.3 Columnwise constrained minimax problem

For column g_j , the exact-leaf linear program is

$$\begin{aligned} u_j(X; G) = \min_{\beta_j, \eta_j} \quad & \eta_j \\ \text{subject to} \quad & -\eta_j \mathbf{1} \leq g_j - X\beta_j \leq \eta_j \mathbf{1}, \\ & -\omega(X) \leq X\beta_j \leq \omega(X), \\ & \eta_j \geq 0. \end{aligned} \quad (34)$$

The leaf score and full score are

$$U_X(G) = \max_{j \in \{01,10,11\}} u_j(X; G), \quad U_{\text{full}}(G) = \min_{X \in \mathcal{X}} U_X(G). \quad (35)$$

Thus $U_{\text{full}}(G)$ is the smallest possible maximum cell discrepancy across the three topology coordinates under the authorized support class and its envelope restrictions.

The score is a constrained topology-discrepancy statistic. It is not a likelihood, posterior, causal effect, performance index, or probability that a mechanism occurred.

3.4 Interval-valued numerical certification

The numerical procedure returns certified lower and upper bounds under the declared optimization tolerances. For Ida,

$$U_{\text{full}}(G_{\text{Ida}}) \in [0.250734484671, 0.250734494671]. \quad (36)$$

The interval width is approximately 10^{-8} . The interval, rather than its midpoint, is used for conservative comparison with reference windows. The primary locked interval is $[0.25073448467090204, 0.25073449467090003]$ and the independent HiGHS replication returns $[0.25073448467103193, 0.2507344946710318]$. The intervals overlap and both support the rounded interval in equation (36). This is numerical replication under stated tolerances, not an exact rational primal–dual certificate.

The value 0.25 is the frozen M7A decision threshold for excluding the full-combined restricted-support class. It is neither a police-performance standard nor an empirical rarity cutoff. M7B ranks Ida using the continuous score intervals for Ida and the qualified reference windows; the rank and interval separation do not depend on classifying observations above or below 0.25.

3.5 Geometric interpretation

For fixed X , equation (34) finds the best approximation to each column of G within the span of X and within its rowwise envelope. The outer minimization searches the authorized finite support library. The statistic therefore combines two ideas:

1. *shape compatibility*: can the support span reproduce the multivariate time path?;
2. *envelope feasibility*: can it do so without exceeding its allowed rowwise support?

A large residual can arise from an unsupported time bin, incompatible cross-coordinate shape, or binding envelope restrictions. It cannot by itself label the omitted factor.

4 Reference-window construction and empirical rarity

4.1 Same-estimator reference principle

Each qualified reference window is processed using the identical ten-bin construction, common-support standardization, topology matrix, support library, and optimization program. Let

$$I_{\text{Ida}} = [L_{\text{Ida}}, U_{\text{Ida}}], \quad I_r = [L_r, U_r] \quad (37)$$

be the Ida and reference score intervals.

A reference is definitely at least as extreme as Ida if $L_r \geq U_{\text{Ida}}$ and possibly at least as extreme if $U_r \geq L_{\text{Ida}}$. Conservative rank bounds are therefore

$$r_L = 1 + \sum_{r=1}^R \mathbf{1}\{L_r \geq U_{\text{Ida}}\}, \quad (38)$$

$$r_U = 1 + \sum_{r=1}^R \mathbf{1}\{U_r \geq L_{\text{Ida}}\}. \quad (39)$$

The add-one reference-fraction interval is

$$\left[\frac{r_L}{R+1}, \frac{r_U}{R+1} \right]. \quad (40)$$

No reference interval overlaps Ida, so $r_L = r_U = 1$ in all three designs.

Table 2. Locked reference results

Reference design	Windows R	Ida rank	Add-one fraction
Full qualified	217	1	$1/218 = 0.00459$
Stage-era matched	153	1	$1/154 = 0.00649$
Same-season stage	86	1	$1/87 = 0.01149$

4.2 Why these fractions are not causal p-values

The design does not impose random assignment of Hurricane Ida across candidate windows and does not adopt an exchangeability assumption. The reference fractions answer a descriptive question:

Where does Ida’s topology-discrepancy score lie relative to the qualified empirical reference windows processed by the same estimator?

They do not answer a randomization-inference question and do not estimate a no-Ida causal counterfactual.

4.3 Simultaneous cell threshold

For reference window r , define its maximum absolute cell contrast

$$M_r = \max_{t \in \{1, \dots, 10\}, j \in \{01, 10, 11\}} |\Delta_{j,t}^{(r)}|. \quad (41)$$

The common threshold is the frozen inverse-CDF 95th percentile of $\{M_r\}_{r=1}^{217}$:

$$c_{0.95} = 0.209397476359. \quad (42)$$

A focal cell is called abnormal when $|\Delta_{j,t}| > c_{0.95}$. This is a familywise empirical threshold across the thirty primary cells, not thirty independent tests.

The nine threshold-exceeding cells are listed in [table 3](#).

Table 3. Nine threshold-exceeding Ida cells with calendar labels

Bin	Calendar interval	State	Contrast
B2	29 Aug. PM	J_{11}	-0.422
B3	30 Aug. AM	J_{11}	-0.354
B4	30 Aug. PM	J_{11}	-0.216
B5	31 Aug. AM	J_{11}	-0.266
B6	31 Aug. PM	J_{01}	+0.307
B6	31 Aug. PM	J_{11}	-0.507
B7	1 Sep. AM	J_{01}	+0.323
B7	1 Sep. AM	J_{11}	-0.455
B8	1 Sep. PM	J_{11}	-0.228

The phrase “two-phase topology” is only a compact description of this set-valued pattern:

$$\mathcal{A}_{\text{early}} = \{(t, 11) : t \in \{B2, B3, B4, B5\}\}, \quad (43)$$

$$\mathcal{A}_{\text{late}} = \{(t, 11), (t, 01) : t \in \{B6, B7\}\}. \quad (44)$$

It is not a fitted two-state process, a statistically estimated change point, or evidence of two causal mechanisms.

5 Within-disposition localization and exact accounting

5.1 Joint masses and conditional public shares

For recorded disposition r , population z , and bin t , let

$$H_{01,r,t}^z = \Pr(J = 01, R = r \mid z, t), \quad (45)$$

$$H_{11,r,t}^z = \Pr(J = 11, R = r \mid z, t), \quad (46)$$

where the probabilities are the standardized public masses under the inherited parent estimator. Define the arrival-observed mass

$$m_{r,t}^z = H_{01,r,t}^z + H_{11,r,t}^z. \quad (47)$$

When $m_{r,t}^z > 0$, define

$$q_{r,t}^z = \frac{H_{11,r,t}^z}{m_{r,t}^z} = \Pr(D = 1 \mid A = 1, R = r, z, t). \quad (48)$$

This is the public dispatch-field share among arrival-observed records within disposition r . It is not a physical dispatch probability.

The identities

$$H_{11,r,t}^z = m_{r,t}^z q_{r,t}^z, \quad H_{01,r,t}^z = m_{r,t}^z (1 - q_{r,t}^z) \quad (49)$$

separate disposition mass from the conditional public field relationship.

5.2 Exact symmetric Kitagawa identity within a disposition

For any two products $m^E q^E$ and $m^R q^R$,

$$m^E q^E - m^R q^R = \frac{q^E + q^R}{2}(m^E - m^R) + \frac{m^E + m^R}{2}(q^E - q^R). \quad (50)$$

Proof. Expand the right-hand side:

$$\begin{aligned} & \frac{1}{2}(q^E m^E - q^E m^R + q^R m^E - q^R m^R) \\ & + \frac{1}{2}(m^E q^E - m^E q^R + m^R q^E - m^R q^R) \\ & = m^E q^E - m^R q^R. \end{aligned}$$

□

Applied to H_{11} , the first term is the disposition-mass component and the second is the within-disposition public-field component. For $H_{01} = m(1 - q)$,

$$\begin{aligned} \Delta H_{01,r,t} &= \frac{2 - q_{r,t}^E - q_{r,t}^R}{2}(m_{r,t}^E - m_{r,t}^R) \\ &\quad - \frac{m_{r,t}^E + m_{r,t}^R}{2}(q_{r,t}^E - q_{r,t}^R). \end{aligned} \quad (51)$$

The same conditional-share change enters H_{11} and H_{01} with opposite signs.

5.3 Composition-only exclusion theorem

Theorem 1: composition-only sign restriction

Fix r and t . Under the restricted composition-only model

$$q_{r,t}^E = q_{r,t}^R = q_{r,t}, \quad (52)$$

we have

$$\Delta H_{11,r,t} = q_{r,t} \Delta m_{r,t}, \quad \Delta H_{01,r,t} = (1 - q_{r,t}) \Delta m_{r,t}, \quad (53)$$

and therefore

$$\Delta H_{11,r,t} \Delta H_{01,r,t} = q_{r,t} (1 - q_{r,t}) (\Delta m_{r,t})^2 \geq 0. \quad (54)$$

Thus opposite-signed $\Delta H_{11,r,t}$ and $\Delta H_{01,r,t}$ exclude composition-only in that cell.

This result requires no stochastic test and no choice among decomposition orders. It is a finite-file accounting implication.

5.4 Six independent sign witnesses

The focal GOA, RTF, and NAT cells have the joint contrasts in table 4.

Table 4. Joint public masses in the six focal disposition–time cells

Calendar bin	Disposition	$\Delta H_{01,r}$	$\Delta H_{11,r}$
31 Aug. PM	GOA	+0.119667	−0.074103
31 Aug. PM	RTF	+0.052795	−0.201984
31 Aug. PM	NAT	+0.135677	−0.220860
1 Sep. AM	GOA	+0.060965	−0.023359
1 Sep. AM	RTF	+0.063914	−0.192636
1 Sep. AM	NAT	+0.192744	−0.220643

All six products $\Delta H_{01,r}\Delta H_{11,r}$ are negative, so composition-only is excluded in every focal cell. The contemporaneous disposition registry defines GOA as Gone on Arrival, RTF as Report to Follow, NAT as Necessary Action Taken, and UNF as Unfounded. The 2021 accounting is qualified; an unchanged cross-year disposition semantic chain is not assumed.

5.5 Aggregate decomposition across dispositions

Among arrival-observed records, define the disposition share

$$\pi_{r,t}^z = \Pr(R = r \mid A = 1, z, t), \quad (55)$$

and aggregate public dispatch-field share

$$q_{\text{all},t}^z = \sum_r \pi_{r,t}^z q_{r,t}^z. \quad (56)$$

Adding and subtracting symmetric cross-products yields

$$\begin{aligned} q_{\text{all},t}^{\text{E}} - q_{\text{all},t}^{\text{R}} &= \sum_r \frac{q_{r,t}^{\text{E}} + q_{r,t}^{\text{R}}}{2} (\pi_{r,t}^{\text{E}} - \pi_{r,t}^{\text{R}}) \\ &\quad + \sum_r \frac{\pi_{r,t}^{\text{E}} + \pi_{r,t}^{\text{R}}}{2} (q_{r,t}^{\text{E}} - q_{r,t}^{\text{R}}). \end{aligned} \quad (57)$$

The first sum is the composition component; the second is the within-disposition component.

Table 5. Locked aggregate decomposition

Calendar bin	Total	Composition	Within disposition
31 Aug. PM	−0.4950345505	+0.0118915647	−0.5069261153
1 Sep. AM	−0.4823594957	+0.0122764739	−0.4946359695

The maximum identity residual is below 1.7×10^{-16} . The within term has the same sign as the total and a much larger absolute magnitude than the composition term.

5.6 Order sensitivity

Twofold decompositions can also be written sequentially by using event weights or reference weights. For 31 August PM, the two extreme orderings are approximately

$$-0.49503 = (+0.00426) + (-0.49929) \quad (58)$$

or

$$-0.49503 = (+0.01952) + (-0.51456). \quad (59)$$

For 1 September AM they are

$$-0.48236 = (+0.00245) + (-0.48481) \quad (60)$$

or

$$-0.48236 = (+0.02210) + (-0.50446). \quad (61)$$

The symmetric values lie between these extremes. Therefore the conclusion that the within-disposition component dominates is not an artifact of one arbitrary ordering.

6 Measure-specific structural partial identification

The results in this section instantiate [equation \(3\)](#) for the operational and DV measures used in the paper. Each subsection fixes the relevant denominator before deriving the identified set.

6.1 Conditional internal-stage/public-capture bridge

Let D_{pub} be the public dispatch-field indicator and let U_{dispatch} be a candidate qualified internal administrative dispatch-event indicator. Conditional on $A_{\text{pub}} = 1$, recorded disposition r , and bin t , define

$$q = \Pr(D_{\text{pub}} = 1), \quad (62)$$

$$u = \Pr(U_{\text{dispatch}} = 1), \quad (63)$$

$$\chi = \Pr(D_{\text{pub}} = 1 \mid U_{\text{dispatch}} = 1). \quad (64)$$

If semantic evidence and the quantitative defect gate establish exact nesting

$$D_{\text{pub}} \leq U_{\text{dispatch}}, \quad (65)$$

then

$$q = u\chi. \quad (66)$$

Theorem 2: product non-identification

For $0 < q < 1$,

$$\mathcal{I}(u, \chi \mid q) = \{(u, q/u) : u \in [q, 1]\}. \quad (67)$$

The set contains a continuum of observationally equivalent factor pairs. At $q = 1$, $u = \chi = 1$. At $q = 0$, if $u > 0$ then $\chi = 0$; if $u = 0$, χ is not defined in the cell because its conditioning event has zero mass.

Proof. Equation (66) implies $u \geq q$ and $\chi = q/u$. Every $u \in [q, 1]$ produces a feasible $\chi \in [q, 1]$. Conversely, every feasible pair must lie on this curve. The boundary cases follow directly. $\square$

The bridge is itself not qualified in the locked public 2021 data. Therefore the unconditional structural uncertainty is no smaller than the conditional product set.

6.2 Identification from joint counts

If a direct bridge is admitted and valid joint counts are available, define

$$N_{du} = \#\{D_{\text{pub}} = d, U_{\text{dispatch}} = u\}, \quad d, u \in \{0, 1\}. \quad (68)$$

Under exact nesting, $N_{10} = 0$. Let

$$N = N_{00} + N_{01} + N_{10} + N_{11}. \quad (69)$$

Then

$$u = \frac{N_{01} + N_{11}}{N}, \quad \chi = \frac{N_{11}}{N_{01} + N_{11}}, \quad (70)$$

when denominators are positive.

The bridge-defect estimand is

$$\delta = \Pr(D_{\text{pub}} = 1, U_{\text{dispatch}} = 0). \quad (71)$$

If semantics imply equation (65) but the suppression-aware upper bound on N_{10} is positive, the capture factorization is not admitted.

If only marginals $d = \Pr(D_{\text{pub}} = 1)$ and $u = \Pr(U_{\text{dispatch}} = 1)$ are available, the overlap quantity

$$\kappa_{DU} = \Pr(D_{\text{pub}} = 1 \mid U_{\text{dispatch}} = 1) \quad (72)$$

has the sharp Fréchet bounds, for $u > 0$,

$$\frac{\max(0, d + u - 1)}{u} \leq \kappa_{DU} \leq \frac{\min(d, u)}{u}. \quad (73)$$

Sharpness here is within the class of normalized nonnegative 2×2 tables with fixed marginals d and u : both overlap endpoints are attained by feasible tables. It does not establish that the variables form a qualified institutional bridge. It is called capture only after direct-bridge semantics and nesting are admitted.

6.3 Queue non-identification

Queue quantities live on the created-call universe, not the arrival-observed bridge universe. Let

$$Q_{t+1} = Q_t + I_t - E_t^{\text{asg}} - E_t^{\text{can}} - E_t^{\text{alt}}. \quad (74)$$

Theorem 3: bridge information does not identify queue state

For any admissible internal-stage prevalence u_{dispatch} , the two stock-flow paths

$$(Q_t, I_t, E_t^{\text{asg}}, E_t^{\text{can}}, E_t^{\text{alt}}, Q_{t+1}) = (0, a, a, 0, 0, 0) \quad (75)$$

and

$$(Q_t, I_t, E_t^{\text{asg}}, E_t^{\text{can}}, E_t^{\text{alt}}, Q_{t+1}) = (b, a, a, 0, 0, b) \quad (76)$$

are both feasible for $a, b > 0$ and share the same bridge observation. Hence bridge prevalence alone identifies neither queue stock, realized availability, cancellation, alternate exits, nor physical performance.

6.4 Interval queue bounds and infeasibility

Suppose $Q_t \in [Q_L, Q_U]$, $I_t \in [I_L, I_U]$, and $Q_{t+1} \in [Q_L^+, Q_U^+]$. Let total nonnegative exit be $E_t = Q_t + I_t - Q_{t+1}$. If

$$Q_U + I_U < Q_L^+, \quad (77)$$

then the stock-flow set is empty:

$$\text{EMPTY_STOCK_FLOW_SET}. \quad (78)$$

Otherwise a certified outer interval is

$$E_t \in \left[\max(0, Q_L + I_L - Q_U^+), Q_U + I_U - Q_L^+ \right]. \quad (79)$$

It is sharp only when the endpoints are jointly attainable under every other constraint.

6.5 Priority endpoints do not identify priority transport

Let P^0 and P^r be initial and recorded priority endpoints. Their joint distribution identifies endpoint disagreement. It does not identify an ordered history

$$P^0 \rightarrow P^{(1)} \rightarrow \dots \rightarrow P^r, \quad (80)$$

its event times, actor classes, reasons, or overwritten intermediate states. Even equality $P^0 = P^r$ does not prove stability; a call could move away from and later return to its initial value.

Let $H = 1$ indicate that at least one priority change occurred. Endpoint disagreement is sufficient for $H = 1$, so without further history the sharp prevalence bound is

$$\Pr(P^0 \neq P^r) \leq \Pr(H = 1) \leq 1. \quad (81)$$

The lower endpoint assigns no hidden changes to equal-endpoint records; the upper endpoint assigns at least one offsetting sequence to every such record. Both completions preserve the observed endpoints.

A qualified transport matrix requires a common risk set and joint transitions

$$\Pi_{ab,t} = \Pr(P^{\text{post}} = b \mid P^{\text{pre}} = a, t), \quad (82)$$

not separate endpoint marginals.

6.6 Continuity semantics versus realized states

Policy can define possible fallback routes without identifying which route was realized. A component state should be represented as a product

$$S_{c,t} = (C_{c,t}, F_{c \rightarrow c',t}), \quad (83)$$

where

$$C_{c,t} \in \{\text{NORMAL}, \text{DEGRADED}, \text{UNAVAILABLE}, \text{UNKNOWN}\} \quad (84)$$

and

$$F_{c \rightarrow c',t} \in \{\text{ACTIVE}, \text{INACTIVE}, \text{UNKNOWN}\}. \quad (85)$$

This allows the scientifically important state “primary unavailable, fallback active”. A single binary outage flag would collapse these cases.

6.7 Regularization is not identification

Let the data-consistent structural set be

$$\Theta_I(Y) = \{\theta : H(\theta) = Y, \theta \in \Theta\}. \quad (86)$$

A regularized selector

$$\theta^* = \arg \min_{\theta \in \Theta_I(Y)} R(\theta) \quad (87)$$

may be unique even when $\Theta_I(Y)$ contains many points. Uniqueness of θ^* identifies the minimizer of the chosen regularizer, not the historical latent state. This distinction is essential before any minimum-action or sparsity-based pathway search.

6.8 Declared-rule DV labeling bounds

Let $\mathcal{L}$ be a declared administrative labeling rule and let

$$C_i^{\mathcal{L}} \in \{1, 0, U\} \quad (88)$$

denote that record i is labeled DV-related, labeled other, or unresolved under $\mathcal{L}$. These are outputs of the declared administrative rule, and U records unresolved classification authority. Define

$$N_1 = \sum_i \mathbf{1}\{C_i^{\mathcal{L}} = 1\}, \quad N_U = \sum_i \mathbf{1}\{C_i^{\mathcal{L}} = U\}. \quad (89)$$

For each unresolved record let $u_i \in \{0, 1\}$ denote an admissible completion under the rule. Then

$$N_{\text{DV}}^{\mathcal{L}} = N_1 + \sum_{i: C_i^{\mathcal{L}} = U} u_i \in [N_1, N_1 + N_U]. \quad (90)$$

The bound is sharp over rule-consistent completions: assigning all unresolved records to 0 attains the lower endpoint, assigning all to 1 attains the upper endpoint, and choosing any intermediate number attains every integer between them. For a fixed CAD denominator N , division by N

yields the corresponding share bound. The completion logic matches sharp binary missing-outcome bounds (Manski, 2003). Auxiliary restrictions on actual misclassification probabilities could tighten a different identified set (Molinari, 2008). Imperfect verification can similarly yield informative performance bounds when the verified and unverified groups are explicitly modeled (Dominitz and Sherman, 2006). Priority, endpoint, EPR, DVU, and AIR stages retain their own denominators and links.

6.9 Endpoint-selected response-time distributions

Fix a declared labeling rule and condition on $C^{\mathcal{L}} = 1$. Let $E = 1$ indicate that both endpoints required by the declared response-time clock are observed. Define

$$\pi_E = \Pr(E = 1 \mid C^{\mathcal{L}} = 1), \quad F_E(a) = \Pr(\tau \leq a \mid C^{\mathcal{L}} = 1, E = 1). \quad (91)$$

The selected distribution F_E is a function of the endpoint-observed records. The full-denominator distribution satisfies

$$F_{\tau}(a) = \pi_E F_E(a) + (1 - \pi_E) F_0(a), \quad (92)$$

where $F_0(a) = \Pr(\tau \leq a \mid C^{\mathcal{L}} = 1, E = 0)$ is unobserved. Hence

$$\pi_E F_E(a) \leq F_{\tau}(a) \leq \pi_E F_E(a) + (1 - \pi_E). \quad (93)$$

Sharpness. At a fixed threshold a , assigning all endpoint-missing values above a attains the lower endpoint, while assigning all below or at a attains the upper endpoint. Every intermediate value is attained by assigning the corresponding fraction to each side. The bounds therefore require no assumption about the endpoint-missing distribution. A qualified classifier must be incorporated before these bounds are interpreted for a strict DV population.

6.10 Lineage-selected administrative follow-through

Again condition on $C^{\mathcal{L}} = 1$. Let $L = 1$ indicate retained lineage from the CAD denominator to a declared follow-through stage, and let $Z = 1$ indicate completion of the specified duty. Write

$$\pi_L = \Pr(L = 1 \mid C^{\mathcal{L}} = 1), \quad p_L = \Pr(Z = 1 \mid C^{\mathcal{L}} = 1, L = 1). \quad (94)$$

Then

$$p_Z = \pi_L p_L + (1 - \pi_L) p_0, \quad p_0 \in [0, 1], \quad (95)$$

and the sharp bounds are

$$\pi_L p_L \leq p_Z \leq \pi_L p_L + (1 - \pi_L). \quad (96)$$

The endpoint assignments $p_0 = 0$ and $p_0 = 1$ attain the two limits. If π_L is unavailable, the current-data set for p_Z is $[0, 1]$. Aggregate retained lineage identifies π_L ; linked completion counts identify p_L .

7 Prospective technical frontier: model uncertainty and reachability

This section preserves the prospective machinery for future internal-witness extensions. The current empirical and DV measurement results require only Sections 1–6; the 54-regime construction below records how stronger evidence would open additional structural objects.

7.1 Tagged union over model classes

Let $m \in \mathfrak{M}$ index a registered structural model class. Its class-specific identified set is

$$\Theta_I^m(Y, W) = \{(m, \theta_m, X_{1:T}) : Y_t \in H_m(X_t; \theta_m), X_{t+1} \in F_m(X_t, a_t; \theta_m), \mathcal{H}_m(W)\}, \quad (97)$$

where W denotes the admitted witness regime and $\mathcal{H}_m(W)$ collects the hard semantic, retention, denominator, suppression, and accounting restrictions.

The model-uncertainty object is the tagged union

$$\Theta_I^\cup(Y, W) = \bigcup_{m \in \mathfrak{M}(W)} \{m\} \times \Theta_I^m(Y, W). \quad (98)$$

The class tag is retained so that similarly named coordinates are not silently equated across models.

The registered templates include:

- a public accounting baseline;
- a restricted composition-only diagnostic;
- an internal/public bridge product;
- queue and realized-availability correspondence;
- priority-risk-set and exit transport;
- component continuity and fallback;
- integrated B-Q, B-Q-P, and B-Q-P-C classes under explicit coupling maps.

Legal uncoupled module combinations are represented by Cartesian products. An integrated class denotes an additional cross-module restriction, not merely joint availability of the component data.

7.2 Reachability and viability

For class m , let $\mathcal{R}_1^m$ be the reference-qualified initial state set. Recursively define the reachable set

$$\mathcal{R}_{t+1}^m = \{x' : \exists x \in \mathcal{R}_t^m, \exists a_t \in \mathcal{A}_{m,t} \text{ such that } x' \in F_m(x, a_t)\}. \quad (99)$$

The observation-compatible viability set is

$$\mathcal{V}_t^m = \mathcal{R}_t^m \cap H_m^{-1}(\mathcal{Y}_{\text{Ida},t}). \quad (100)$$

The linked path set is

$$\mathcal{X}_I^m = \{X_{1:T} : X_t \in \mathcal{V}_t^m, X_{t+1} \in F_m(X_t, a_t)\}. \quad (101)$$

Algebraic fit without a full linked path is not admissible.

For latent coordinate $x_{g,t}$, the sharp bounds are

$$\underline{x}_{g,t}^m = \inf_{X \in \mathcal{X}_I^m} x_{g,t}, \quad \bar{x}_{g,t}^m = \sup_{X \in \mathcal{X}_I^m} x_{g,t}. \quad (102)$$

A bound is called sharp only when endpoints are attainable or arbitrarily approximable by feasible paths. Otherwise it is labeled `OUTER_NOT_SHARP`.

7.3 Current W_0 boundary

At the current pre-Stage-S/pre-Stage-V witness state W_0 , only two public classes are open:

1. the observation baseline, which is sharp for qualified public coordinates;
2. the restricted composition-only diagnostic, which is empty in the six focal late-phase cells.

No latent structural class is admitted. Consequently, bounds and possible/required sets for internal dispatch, public capture, queue pressure, realized availability, priority transport, alternate exits, and continuity are `NOT_DEFINED`. This is a certification of the identification boundary, not a failure to optimize.

7.4 Witness-availability phase diagram

Four modules are considered:

$$B = \text{internal/public bridge}, \quad Q = \text{queue and availability}, \quad (103)$$

$$P = \text{priority transport}, \quad C = \text{continuity/fallback}. \quad (104)$$

Each can be `CLOSED`, `PARTIAL`, or `QUALIFIED`. Encode those levels as 0, 1, and 2. The queue module has $Q \in \{0, 1, 2\}$ and the priority module must satisfy $P \in \{0, \dots, Q\}$, producing the six ordered pairs

$$(Q, P) \in \{(0, 0), (1, 0), (1, 1), (2, 0), (2, 1), (2, 2)\}. \quad (105)$$

With three independent levels for each of B and C , the number of legal witness-availability regimes is

$$3_B \times 3_C \times 6_{(Q,P)} = 54. \quad (106)$$

The 54 regimes do not exhaust every possible coupling model. Every multi-module regime separately records whether cross-module propagation is

$$\text{UNCOUPLED}, \quad \text{PARTIAL_MAP}, \quad \text{QUALIFIED_MAP}. \quad (107)$$

Joint observation of two modules does not identify transmission between them.

7.5 Eight minimum-witness objects

Table 6. Minimum evidence for the eight frozen future identification objects

Object	Minimum witness configuration
A. Bridge-factor separation	qualified B, exact nesting, valid joint $D_{\text{pub}} \times U_{\text{dispatch}}$ counts
B1. Queue path	event-period queue stock on a qualified risk set
B2. Queue abnormality	B1 plus same-semantic reference queue path and frozen reference design
C. Queue versus realized availability	qualified queue stock plus separately defined realized availability
D1. Priority trajectory	qualified Q and P plus priority-specific exits
D2. Priority-to-public propagation	D1 plus qualified destination B and an admitted $\Gamma_{P \rightarrow B}$ map
E1. Continuity-to-queue propagation	qualified C and Q plus an admitted $\Gamma_{C \rightarrow Q}$ map
E2. Continuity-to-bridge propagation	qualified C and B plus an admitted $\Gamma_{C \rightarrow B}$ map

Even the richest witness regime leaves model probabilities, pathway probabilities, and the historical minimum-action path unidentified unless an empirically calibrated stochastic model is added and passes a separate observability and prior-sensitivity gate.

8 Current-public-data measurement-regime replay

8.1 Question and source universe

The replay asks: if the locked Ida administrative signature were observed under the 2025–2026 public measurement architecture, would the mechanism-relevant identified set become smaller? It does not ask how police would behave if Ida occurred today.

The official frozen sources include the 2025 and 2026 annual CFS datasets and dictionaries, the public NOPD dashboard directory, the response-time Power BI semantic model and aggregate queries, current response and alternative-response policies, the 2024 Sustainment Plan, and 2025 EPR metadata.

8.2 Raw schema genealogy

The same core sixteen public fields are present in 2021, 2025, and 2026. The 2025 and 2026 attached CFS dictionaries are byte-identical. The architecture remains endpoint-based: initial and recorded signal/priority, create/dispatch/arrival/closure timestamps, final disposition, and self-initiation.

The dashboard adds candidate enrichments, including:

- Incident_To_EnRoute_Seconds;
- All Dispositions;
- initial-versus-recorded priority and signal change flags;
- modifier and source-detail categories.

These are public and machine-queryable, but the current public documentation does not establish their construction, ordering, clocks, multiplicity, overwrite behavior, or a binding to the 2021 Ida estimand.

8.3 Conditional semantic-invariance result

Let Y_{26}^{raw} denote the normalized common endpoints in the current annual schema, and let $D_{26}^{\text{derived}} = g(Y_{26}^{\text{raw}})$ be a deterministic dashboard transformation.

Theorem 4: deterministic public transforms do not contract their raw-input identified set

If $D_{26}^{\text{derived}} = g(Y_{26}^{\text{raw}})$ is deterministic, then for any parameter θ defined under semantic map h ,

$$\mathcal{I}_{2026}(\theta \mid Y_{\text{Ida}}, Y_{26}^{\text{raw}}, D_{26}^{\text{derived}}, h) = \mathcal{I}_{2026}(\theta \mid Y_{\text{Ida}}, Y_{26}^{\text{raw}}, h). \quad (108)$$

Proof. Because D_{26}^{derived} is a deterministic function of Y_{26}^{raw} , every feasible world consistent with the raw endpoints already fixes the derived value. Adding that value leaves the compatibility set, and therefore its image in θ , unchanged. $\square$

The theorem is within the 2026 information regime. It does not equate the 2026 and 2021 identified sets; that stronger claim would require equality of the restrictions defining feasible worlds across the two regimes. Dashboard-only fields also require a qualified semantic binding. For queue, realized availability, priority-event history, applicable callback/fallback states, and component

continuity, the decisive state variables remain absent. No strict common-estimand contraction is proved.

8.4 Public semantic enrichment versus structural witness qualification

The replay distinguishes public descriptive enrichment from realized M8 structural witnesses:

Table 7. Semantic enrichment does not open the latent structural modules

Module	Current public enrichment	Realized M8 witness status
B: internal/public dispatch bridge	partial context; separate public surfaces and a 2025 CAD–EPR item link	closed
Q: queue and realized availability	endpoint and duration proxies only	closed
P: priority transport	endpoint enriched; policy and change flags	closed
C: continuity and fallback	semantics documented in policy	closed

A CAD–EPR item link is not the internal-dispatch/public-dispatch bridge. Response durations do not identify queue stock. Priority endpoint disagreement is not an event history. Policy descriptions of fallback do not reveal realized continuity states. The realized M8 witness regime therefore remains W_0 .

8.5 Current public-data validity audit

Table 8. Selected 2025–2026 public-data validity results

Check	2025	2026 snapshot
Rows	329,770	209,829
Unique item IDs	329,770	209,829
Dispatch field present	41.80%	41.88%
Arrival field present	83.72%	84.14%
Arrival present, dispatch absent	152,701	97,573
Malformed nonblank stage timestamps	0	0
Exact priority-endpoint disagreement	38.74%	37.10%
Numeric-root priority disagreement	23.32%	24.16%

These are properties of public administrative files. They do not measure physical response, effective capacity, or police performance. The high prevalence of arrival-present/dispatch-absent records in contemporary data reinforces why this configuration cannot be interpreted mechanically as nonresponse.

8.6 Replay decision

The supported decision is

$$\boxed{\text{M8P_PUBLIC_OBSERVABILITY_PARTIALLY_IMPROVED.}} \quad (109)$$

The raw architecture remains substantially the same, while the dashboard and current policies add useful descriptive and semantic layers. No mechanism-relevant strict identified-set contraction is proved.

9 Statistical and econometric interpretation

9.1 Finite-population exactness

The reported topology contrasts and decomposition identities are exact functionals of the frozen public data and estimator. Numerical tolerances arise from floating-point computation and certified optimization, not from sampling the finite file. This finite-population exactness should be distinguished from superpopulation consistency or causal inference.

9.2 Reference uncertainty

The reference rank summarizes empirical rarity across qualified windows. It is robust to interval-valued optimization because every reference interval is separated from *Ida*. The ten bins form one multivariate window, and the finite-file decomposition is exact.

If future work seeks uncertainty for a partially identified structural parameter, the appropriate object is a confidence region for the observable moments propagated through the identified-set map. If $C_{1-\alpha}$ is a simultaneous confidence set for observable moments m_0 , then

$$\mathcal{C}_{1-\alpha}^\theta = \bigcup_{m \in C_{1-\alpha}} \mathcal{I}(\theta | m) \quad (110)$$

has at least $1 - \alpha$ coverage whenever $m_0 \in C_{1-\alpha}$. Plugging point estimates into a linear program and attaching an ordinary regression standard error afterward is not valid set inference.

9.3 When minimum-action analysis becomes meaningful

Once qualified latent witnesses enter, a reference-calibrated action could be defined as

$$\mathcal{A}(z) = \sum_{t=1}^T z_t' \widehat{\Sigma}_e^{-1} z_t, \quad (111)$$

where z_t is an *Ida* deviation from a normal transition law estimated only from reference windows. The least-reference-deviation path would be

$$z^* = \arg \min_{z \in \mathcal{P}_I} \mathcal{A}(z). \quad (112)$$

Before a stochastic model is admitted, this remains a selector from a feasible path set. It is not a historical probability statement.

9.4 Stochastic state-space and Bayesian layer

A probabilistic pathway analysis requires transition and observation laws such as

$$X_{t+1} \sim p_\theta(X_{t+1} | X_t), \quad (113)$$

$$Y_t \sim p_\theta(Y_t | X_t). \quad (114)$$

Only after semantic qualification, structural identification/observability checks, reference out-of-sample validation, and prior-sensitivity analysis would it be legitimate to report posterior pathway mass or a maximum a posteriori path. No such probabilities are produced in the current study.

10 Reproduction guide

10.1 Reproduction principle

All focal numbers are bound to locked aggregate artifacts. Replication scripts should be run in a disposable copy of the repository because some verification programs write result files beside their inputs. The main estimators need not be rerun to verify the paper; the locked machine outputs and independent replication reports are sufficient.

10.2 Primary result map

All paths below are repository-relative. The M7B/M7D-E files live under the locked New Orleans Wave4R candidate tree; the M8P files live under the current-public-data replay tree.

Table 9. Result-to-artifact map

Result	Primary locked artifact	Independent check
Ida topology score and ranks	M7B_IDA_RANK_BOUNDS.json	M7B_INDEPENDENT_REPLICATION.json
Nine abnormal cells	M7B_IDA_ABNORMAL_CELLS.csv	maximum-cell threshold replay
Within-disposition joint masses	M7D_E_RESULTS.json	M7D_E_INDEPENDENT_REPLICATION.json
Kitagawa decomposition	M7D_E_RESULTS.json	aggregate-only implementation
M8 structural boundary	BELAND_PLUS_M8A_LATENT_OPERATIONAL_DYNAMICS.md	M8B-R1, M8C, and M8D lock receipts
Current-data audit	current_data_validity_audit.json	validation_report.json
M8P replay decision	m8p_results.json	M8P-R1 lock receipt

10.3 Expected verification commands

From the repository root, use the governed Python environment and run:

```
python pilot_911_dv/experiments/
nola_2020-01-01_2024-12-31_beland_plus_wave4r/candidate/
m7b_same_estimator_reference_geometry/independent_replicate_m7b.py

python pilot_911_dv/experiments/
nola_2020-01-01_2024-12-31_beland_plus_wave4r/candidate/
m7d_e_within_disposition_dispatch_observability/replicate_m7d_e.py

python pilot_911_dv/experiments/
nola_2025-01-01_2026-08-11_m8p_public_observability_replay/
notebooks/audit_current_public_data.py
```

```
python pilot_911_dv/experiments/
nola_2025-01-01_2026-08-11_m8p_public_observability_replay/
notebooks/build_m8p_outputs.py

python pilot_911_dv/experiments/
nola_2025-01-01_2026-08-11_m8p_public_observability_replay/
notebooks/validate_m8p.py
```

The line wrapping above is typographic. In an actual shell, each script path is supplied as one argument.

10.4 Expected invariants

A valid M7B replay must satisfy:

- 217 full qualified reference windows;
- Ida matrix parity no larger than 10^{-10} ;
- rank 1 in all three reference designs;
- no interval overlap between Ida and any reference.

A valid M7D-E replay must satisfy:

- parent topology aggregation error below 10^{-10} ;
- Kitagawa identity residual below 10^{-12} ;
- the six opposite-sign composition-only witnesses;
- the B6/B7 aggregate values in [table 5](#).

A valid M8P replay must satisfy:

- decision M8P_PUBLIC_OBSERVABILITY_PARTIALLY_IMPROVED;
- empty strict-contraction list;
- all four realized structural modules closed;
- no event-window or shock fishing in the current-data audit;
- no row-level sensitive examples persisted.

10.5 Frozen current-data snapshot hashes

The frozen bindings are:

```
Official 2025 CFS
dataset_id: 4xwx-sfte
rows: 329770
sha256: be8416343d253e2518a16ae007568a1561ee8b511dbdef3d5465956a198ae875

Official 2026 CFS
dataset_id: es9j-6y5d
rows: 209829
sha256: c151ca38199aa53921ad1fe048ee7108f6165e8700ae459070f4c014ce614e17
```

10.6 Algorithmic skeleton

The public topology and decomposition can be reproduced conceptually by the following pseudocode. It omits project-specific file access and privacy-governance code but preserves the mathematical

order of operations.

```

for window in [ida_window] + qualified_reference_windows:
    records = load_frozen_eligible_records(window)

    for bin_id in frozen_12_hour_bins:
        event_or_reference = records.in_bin(bin_id)
        strata = frozen_common_support[bin_id]
        weights = frozen_reference_weights[bin_id]

        for stratum in strata:
            block = event_or_reference.in_stratum(stratum)
            D = valid_public_dispatch_field(block)
            A = valid_public_arrival_field(block)
            J00, J01, J10, J11 = topology_indicators(D, A)
            store_state_shares(block, J00, J01, J10, J11)

        standardized = weighted_average_over_strata(weights)
        assert simplex_identity(standardized)

    G = build_matrix(columns=[Delta_J01, Delta_J10, Delta_J11])
    score_interval = solve_authorized_support_program(G)

# Focal within-disposition decomposition
for bin_id in [B6, B7]:
    for disposition in [GOA, RTF, NAT]:
        H01, H11 = standardized_joint_masses(bin_id, disposition)
        m = H01 + H11
        q = H11 / m
        verify_cell_kitagawa_identity(m, q)

    verify_aggregate_kitagawa_identity(bin_id)

```

10.7 PDF and figure reproduction

The revised paper and appendix are authored in LaTeX using Latin Modern, the scalable canonical Computer Modern family. All charts are generated in TikZ/PGFPlots, so figure labels and mathematics use the same embedded fonts as the body text. A valid release must pass:

- LaTeX compilation with no unresolved references or citations;
- no overfull boxes in the final log;
- embedded-font inspection;
- page-by-page rendering at 200 dpi;
- visual inspection for clipped equations, tables, labels, or broken glyphs.

10.8 Lean formal-verification supplement

An additive Lean 4 package at `pilot_911_dv/formal_verification_r11_1` checks the R11.1 mathematical inventory and the deterministic-transform correction adopted in R11.2 without modifying the locked empirical inputs. Lean 4 and Mathlib are pinned at version 4.33.0. The kernel-checked tranche contains 65 named theorem declarations, with no `sorry`, `admit`, or project-defined axioms.

Three verification classes apply. *Kernel-verified mathematics* consists of the definitions and theorems accepted by Lean under their stated assumptions. *Tested numerical results* consists of the frozen LP and finite-artifact calculations checked by independent computation but not represented by an exact paper-level rational certificate. *Semantic and institutional bindings* connect formal variables to public fields, operational stages, denominators, and source authority; those bindings require human and source review.

Table 10. Formal-verification and computational-check status

Claim family	Status	Boundary
Public states, standardization, decomposition, bridge, Fréchet, and queue results	FORMALLY_ VERIFIED	Exact statements under their declared mathematical assumptions.
Identified-set contraction, witness sufficiency, DV 1/0/ U bounds, selection bounds, and necessity counterexamples	FORMALLY_ VERIFIED	Target and denominator meanings remain part of the semantic binding.
Deterministic-transform redundancy in equation (108)	FORMALLY_ VERIFIED	Within-2026 equality only; no cross-regime equality is claimed.
Fifty-four witness-availability regimes	FORMALLY_ VERIFIED EXHAUSTIVELY_ ENUMERATED	Availability regimes, not estimated models or pathway probabilities.
M7B Ida numerical interval and ranks	TESTED_ONLY	Primary and independent HiGHS intervals overlap under declared tolerances.
Exact-rational checker fixture	CERTIFICATE_ VERIFIED	Checker test only; it is not the paper LP result.
Exact-rational paper LP certificate	NOT_YET_ VERIFIED	No canonical rational LP instance or exact primal–dual vectors are supplied.
Source semantics and institutional interpretation	SEMANTIC_REVIEW_ REQUIRED	Human and source review must bind fields, clocks, denominators, stages, and authority.

From the repository root, the focused verification commands are:

```
cd pilot_911_dv/formal_verification_r11_1
lake -Kjobs=1 build
python certificates/verify_certificates.py
python enumeration/verify_witness_regimes.py
```

The exact declaration-to-source mapping, assumptions, axiom audit, and interpretation boundaries are recorded in CLAIM_MAP.md, AXIOM_AUDIT.md, and VERIFICATION_REPORT.md. A successful build establishes that Lean accepted the stated formal theorems. It does not verify source-data semantics, empirical estimates, causal claims, mechanism probabilities, or the institutional interpretation of public and operational variables.

11 Claim language and interpretation rules

11.1 Claims supported by the mathematics

The locked analysis permits the following statements:

- Ida is reference-extreme in the frozen public-topology discrepancy statistic.
- Nine public bin–state contrasts exceed a common reference threshold.
- The public-state configuration changes after midday on 31 August: J_{01} becomes positive while J_{11} remains negative.
- Recorded-disposition composition alone is excluded in the six focal GOA/RTF/NAT cells.
- The late aggregate shift is dominated by changes within recorded dispositions.
- A performance functional is directly supportable only when its measure-specific identified set is a singleton under the admitted witnesses.
- Endpoint coverage yields sharp pointwise bounds for a full-denominator response-time distribution.
- Retained lineage and completion coverage yield sharp bounds for a full-denominator administrative follow-through rate.
- Initial and recorded priorities provide a sharp lower bound on the prevalence of any priority change.
- Current public data are descriptively and semantically richer, but no strict mechanism-relevant identified-set contraction is proved.
- Additional joint bridge, queue, availability, priority-history, and continuity witnesses would contract specified parts of the identified set.

11.2 Interpretive boundary

The Ida estimand is the topology of released administrative fields. The downstream performance estimands condition on reported calls created in CAD and require the target-specific witnesses defined above. Earlier help-seeking stages, population DV incidence, victim safety, reform effects, and mechanism probabilities require their corresponding access, outcome, or design evidence. The present evidence supplies no probability ordering among latent pathways.

11.3 Recommended terminology

Preferred terms:

reported calls; reported DV-related calls; public administrative observability; event-window comparison; reference-extreme; within-disposition public field shift; measurement friction; identified set; realized availability; queue stock; recorded disposition; consistent with.

12 Summary of the mathematical logic

The argument has two connected modules:

1. *Released-record stress test.* Define mutually exclusive public field states, standardize event–reference shares, construct $G(Y)$, and compare $T(Y)$ with same-estimator reference windows. The disposition decomposition then localizes the reference-extreme Ida path within recorded

categories.

2. *Measure-specific identification.* Define each candidate target $\theta_k = m_k(S)$ and evaluate $\mathcal{I}_k(Y, W)$. The bridge product, queue identities, priority bounds, $1/0/U$ classification bounds, endpoint-selection bounds, and lineage bounds are applications of this common rule.

The workload map links reported demand, capacity, priorities, and service. The first module shows that instability in the released observation of that system is empirically consequential. The second determines which reported-DV performance functionals remain supportable, bounded, selected, or unavailable and identifies the witnesses that contract each set. Those witnesses define privacy-safe aggregate products for resource management and civilian oversight. The frozen Ida results remain system-level.

References

- Alvarez, F. E., Argente, D., and Van Patten, D. (2026). On the resilience of payment methods. Working Paper 35115, National Bureau of Economic Research.
- Blackwell, D. (1953). Equivalent comparisons of experiments. *Annals of Mathematical Statistics*, 24(2):265–272.
- Carvalho, V. M., Nirei, M., Saito, Y. U., and Tahbaz-Salehi, A. (2021). Supply chain disruptions: Evidence from the great east japan earthquake. *Quarterly Journal of Economics*, 136(2):1255–1321.
- Dominitz, J. and Sherman, R. P. (2006). Identification and estimation of bounds on school performance measures: A nonparametric analysis of a mixture model with verification. *Journal of Applied Econometrics*, 21(8):1295–1326.
- Kapteyn, A. and Ypma, J. Y. (2007). Measurement error and misclassification: A comparison of survey and administrative data. *Journal of Labor Economics*, 25(3):513–551.
- Koopmans, T. C. and Reiersøl, O. (1950). The identification of structural characteristics. *Annals of Mathematical Statistics*, 21(2):165–181.
- Lewbel, A. (2019). The identification zoo: Meanings of identification in econometrics. *Journal of Economic Literature*, 57(4):835–903.
- Manski, C. F. (1989). Anatomy of the selection problem. *Journal of Human Resources*, 24(3):343–360.
- Manski, C. F. (1990). Nonparametric bounds on treatment effects. *American Economic Review*, 80(2):319–323.
- Manski, C. F. (2003). *Partial Identification of Probability Distributions*. Springer, New York.
- Manski, C. F. (2015). Communicating uncertainty in official economic statistics: An appraisal fifty years after morgenstern. *Journal of Economic Literature*, 53(3):631–653.
- Masten, M. A. and Poirier, A. (2020). Inference on breakdown frontiers. *Quantitative Economics*, 11(1):41–111.
- Molinari, F. (2008). Partial identification of probability distributions with misclassified data. *Journal of Econometrics*, 144(1):81–117.

- Molinari, F. (2020). Microeconometrics with partial identification. In *Handbook of Econometrics*, volume 7A, pages 355–486. Elsevier.
- Peters, J., Bühlmann, P., and Meinshausen, N. (2016). Causal inference by using invariant prediction: Identification and confidence intervals. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 78(5):947–1012.
- Rothenberg, T. J. (1971). Identification in parametric models. *Econometrica*, 39(3):577–591.